

Program overview

08-Dec-2022 11:25

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Electrical Engineering

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
B-EE 2022	Bachelor Electrical Engineering 2022		
B-EE jaar 1 2022	Bachelor Electrical Engineering Year 1 2022		
EE1C11	Linear Circuits A	5	
EE1C21	Linear Circuits B	5	
EE1C31	Amplifiers and instrumentation	5	
EE1D11	Digital Systems A	5	
EE1D21	Digital Systems B	5	
EE1L11	EPO-1: Booming Bass	5	
EE1L21	EPO-2: Smart Robot Challenge	5	
EE1M11	Linear Algebra and Analysis A	5	
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EE1M31	Probability and Statistics	5	
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EE2E11	Electrical Energy Conversion	5	
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B-EE 2022

Director of Studies	Dr.ir. N.P. van der Meijs
Program Coordinator	H. Shekarian Tabarestani
Program Coordinator	L.M. Ramirez Elizondo
Program Coordinator	Ir. E.W. Bol
Contact for students	Let op! De antwoorden van veel van je vragen staan al op het web (https://www.tudelft.nl/studenten/faculteiten/ewi-studentenportal/)! Ben je eerstejaars en heb je het antwoord op jouw vraag niet op de website kunnen vinden? Neem dan als eerste contact op met je mentor of je tutor. Voor ouderejaars is de eerste persoon bij wie je terecht kunt de commissaris onderwijs van EE (onderwijs-etv@tudelft.nl). Hij of zij weet zelf op veel vragen antwoord, maar kan ook doorverwijzen of bij andere mensen van de opleiding navragen. Gedurende je hele opleiding worden er regelmatig opleidingsbijeenkomsten gehouden: In het eerste jaar zijn er kwartaalbijeenkomsten, in de hogere jaren semesterbijeenkomsten. De bijeenkomsten staan in het rooster. Voor vragen over hulp en begeleiding bij persoonlijke problemen of studieachterstand kunnen alle studenten contact opnemen met studieadviseurs. Check ook hier eerst de website: https://www.tudelft.nl/studenten/faculteiten/ewi-studentenportal/ Voor antwoord op vragen over specifieke vakken kijk je eerst op de Brightspacesite van dat vak en als je het antwoord daar niet kunt vinden neem je contact op met de verantwoordelijk docent van dat vak. Wil je meedoen aan het verbeteren van de opleiding? Geef je dan op voor de collegeresponsiegroep van jouw jaar. Neem ook hiervoor contact op met je commissaris onderwijs.
Term of validity	Let op! De informatie die je hieronder vindt is informatief van aard en niet bindend! Bindende informatie vind je in de reglementen (https://www.tudelft.nl/studenten/faculteiten/ewi-studentenportal/). Houd Brightspace in de gaten voor wijzigingen en kom naar de opleidingsbijeenkomsten voor de meest recente informatie.
Program Goals	De afgestudeerde van de opleiding Electrical Engineering: - Bezit een brede en grondige kennis van en heeft vaardigheden in de fundamentele ingenieurswetenschappen, die de basis van de elektrotechniek vormen. Hieronder worden verstaan: wiskunde, natuurkunde, informatica en systemen en modellen. De afgestudeerde kan deze kennis actief toepassen op elektrotechnische systemen en beheert de kennis op een zodanig niveau dat toegang verkregen kan worden tot internationaal geaccrediteerde masteropleidingen Electrical Engineering. - Heeft basistechnisch-wetenschappelijke kennis van en heeft vaardigheden in de belangrijkste elektrotechnische disciplines, te weten netwerktheorie, telecommunicatietechniek en systemen, signaalbewerking, elektronische schakelingen, elektrische energietechniek, halfgeleidercomponenten en computertechniek. De afgestudeerde kan deze kennis actief toepassen voor het analyseren en ontwerpen van elektrotechnische systemen en het lezen en begrijpen van de wetenschappelijke literatuur in bovengenoemde vakgebieden. - Bezit basiskennis van en heeft vaardigheden in methodes en gereedschappen voor het modelleren, simuleren, ontwerpen en uitvoeren van experimenten en onderzoek van/aan elektrotechnische systemen. De afgestudeerde kan deze kennis actief toepassen voor het analyseren en synthetiseren van elektrotechnische systemen (op een hoog abstractieniveau). - Kan een bijdrage leveren aan het oplossen van technologische problemen door een systematische wetenschappelijke aanpak. Dit betreft de analyse, het definiëren van innovatieve oplossingen, het onderkennen van de haalbaarheid, het onderkennen en verwerven van ontbrekende kennis, evenals het onderkennen van de betrokkenheid en beperkingen van deze kennis en van de uitwerking van de oplossing. - Kan zowel individueel als in (multidisciplinaire en multinationale) teams werken en waar nodig initiatief nemen. - Kan effectief communiceren (waaronder presenteren en rapporteren) over zijn werk, t.a.v. informatie, problemen, ideeën en oplossingen aan zowel de professionele collegae als aan een niet-specialistisch publiek. - Is in staat om relevante informatie te verzamelen en te interpreteren en kan de technologische, bedrijfskundige, maatschappelijke en ethische gevolgen van zijn werk evalueren en de verantwoordelijkheid nemen met betrekking tot duurzaamheid, economie en sociaal welzijn. De afgestudeerde draagt bij aan de wetenschappelijke praktijk (onderzoeksysteem, relatie met opdrachtgevers, publicatiesysteem, belang van integriteit, etc.). - Toont initiatief en kan kritisch reflecteren (met ondersteuning) op het eigen denken, beslissen, en handelen en dit daarmee bijsturen. De afgestudeerde kan de eigen competenties op peil houden en uitbreiden door permanente zelfstudie, met een hoge mate van zelfstandigheid.
Program Structure 1	Zie het onderwijs- en examenreglement voor de volledige tekst van de eindtermen van de opleiding. De opleiding EE duurt drie jaar. Het eerste en het tweede jaar bestaan elk uit 12 studieonderdelen van elk 5 EC: 10 vakken en twee projecten. Het derde jaar wijkt behoorlijk af. De eerste helft ervan volg je een minor (zie onder). Daarna zijn er nog 3 vakken van 5 EC. Je laatst 15 EC besteed je dat jaar aan je bachelorafstudeerproject. Elk jaar is opgedeeld in 4 kwartalen van 10 weken en tijdens elk kwartaal heb je drie vakken die parallel lopen. Gedurende de 10 weken wordt er van je verwacht dat je elke week ongeveer 40 uur aan je studie besteedt. Drie vakken per kwartaal betekent ongeveer 13 uur per week aan elk van die vakken besteden. Niet al die uren zijn ingepland, je moet veel zelfstandig werken (zelfstudie).
Program Structure 2	EE-studenten en hun docenten maken deel uit van de academische gemeenschap van de TU Delft. Elke gemeenschap kent gedragsregels en de TU Delft heeft deze vastgelegd in een ethische code (Code of Ethics). De onderstaande regels komen uit deze code: Docenten en studenten: - hebben respect voor anderen; - treden betrokken, transparant en integer op; - dragen bij aan een inspirerende werk- en studieomgeving; - vertrouwen elkaar en vermijden belangenverstrengeling; - respecteren elkaars eigendommen en middelen en die van de universiteit. Docenten: - gedragen zich rechtvaardig en respectvol ten opzichte van elkaar en van studenten; - mikken op hoge kwaliteit en streven altijd naar verbetering; Studenten: - stellen zich rechtvaardig en respectvol op ten opzichte van elkaar en van universiteitsmedewerkers; - halen het beste uit zichzelf door actief deel te nemen aan hun opleiding en extra curriculaire activiteiten; - stimuleren elkaar en hun docenten door analytische vragen te stellen en goed beargumenteerde discussies te voeren.
Exam requirements	In de exameneisen staat wanneer je in aanmerking komt voor een EE-diploma. Je vindt ze in de Regels en Richtlijnen van de

	Examencommissie (http://studenten.tudelft.nl/ewi/reglementen/) . Samengevat staat er dat je een EE-diploma krijgt als je alle vakken van de opleiding, inclusief de minor, hebt gehaald met als resultaat een V (voldoende) of een cijfer van 6 of hoger.
With Honours Regulation	Studenten die binnen de voor de opleiding geldende tijd (drie jaar) de BSc opleiding EE afmaken, gemiddeld voor alle vakken minimaal een 8 hebben en ook minimaal een 8 hebben voor het bachelorproject krijgen hun diploma met de aantekening met lof (ook wel: cum laude). De exacte eisen kun je vinden in de Regels en Richtlijnen van de Examencommissie.
Minor Semester	In de eerste helft van je derde jaar volg je een minor. De minor is er om je de mogelijkheid te geven je kennis te verbreden of een buitenlandervaring op te doen. Meer informatie vind je bij het derde jaar.
Gives access to	De bachelor EE geeft o.a. toegang tot de EWI Masters Electrical Engineering, Computer Engineering en Embedded Systems aan de TU Delft. Ook heb je toegang tot de masters met dezelfde naam aan de Universiteit Twente en Technische Universiteit Eindhoven. Met een schakelprogramma kun je toegelaten worden tot een groot aantal andere Masterprogrammas van de TU Delft en andere universiteiten. Zie http://doorstroommatrix.nl/ voor een uitgebreid overzicht.
Expected prior Knowledge	De toelatingseisen van de opleiding EE kun je vinden in hoofdstuk 2 van het Studentenstatuut. Een cijfer van 7 of meer op het VWO diploma voor de vakken Wiskunde en Natuurkunde wordt aanbevolen.
Honours track	Studenten die het eerste academiejaar in één jaar met succes hebben afgerond en de vakken van het eerste academiejaar nominaal en met een gemiddeld cijfer van 8,0 of hoger hebben afgesloten, kunnen zich inschrijven voor het Bachelor Honours Programma (HPB) voor excellente bachelorstudenten. Studenten die het eerste studiejaar nominaal hebben afgerond en een gemiddeld cijfer tussen 7,5 en 8,0 hebben behaald, kunnen een gemotiveerd verzoek indienen om te worden toegelaten. Zie https://www.tudelft.nl/studenten/ewi-studentenportal/onderwijs/honours-programme

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Education	Bachelor Electrical Engineering

B-EE jaar 1 2022	
Director of Studies	Dr.ir. N.P. van der Meijs
Program Coordinator	H. Shekarian Tabarestani
Program Coordinator	Ir. E.W. Bol
Prerequisites	Let op! De Ingangseisen zijn in de OER vastgelegd. Mocht er verschil zijn tussen wat er hier staat en wat in de OER staat, dan is datgene wat in de OER staat geldig. In het eerste jaar gelden voor de volgende twee studieonderdelen ingangseisen: Je mag pas aan het onderdeel (project) EE1L11 EPO-1: De Beukende Bas meedoen als je het practicum van het vak EE1C11 Lineaire Schakelingen A hebt behaald. Je mag pas aan het onderdeel (project) EE1L21 EPO-2: Smart Robot Challenge meedoen als je het practicum van het vak EE1C31 Versterkerschakelingen en Instrumentatie EN het practicum van het vak EE1D21 Digitale Systemen B hebt behaald. Als je niet aan deze ingangseisen voldoet en meent dat je onredelijk veel studieachterstand oploopt als je deze projecten niet mag volgen kun je dit aangeven bij aanmelding voor het project. Mentoraat Het BSc EE Y1 Mentoraatprogramma loopt het hele jaar door. Registratie van dit programma is opgenomen als een pass/fail binnen het EPO-1 (EE1L11) en EPO-2 (EE1L21). Rekenmachine Alleen een eenvoudige rekenmachine zonder mogelijkheden voor opslag van tekst of formules is toegestaan. Geschikte types zijn (varianten van) de Texas Instruments TI-30 en de Casio FX-82.
Introduction 1	Inschrijven Voor de (deel-)tentamens van alle vakken moet je je uiterlijk 1 week vóór de dag van het tentamen via OSIRIS inschrijven. Wil je bij nader inzien toch niet meedoen aan een studieonderdeel of tentamen waar je je voor hebt ingeschreven? Schrijf je dan ook weer uit!
Introduction 2	Zie voor uitleg https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens Zoals bij alle TU Delft bacheloropleidingen, geldt bij EE de norm van 45 EC als bindend studieadvies (BSA). Kijk voor meer informatie op http://bsa.tudelft.nl.

EE1C11	Linear Circuits A	5
Responsible Instructor	Dr.ing. I.E. Lager	
Instructor	Ir. E.W. Bol	
Instructor	Dr. M. Ghaffarian Niasar	
Instructor	B. Abdikivanani	
Instructor	Dr. F. Fioranelli	
Instructor	Dr. I. Ercan	
Contact Hours / Week x/x/x/x	4/0/0/0 hc; 4/0/0/0 pr	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	N.A.	
Expected prior knowledge	N.A.	
Course Contents	This course concerns the SYSTEM analysis of complex, linear circuits. At its core is the calculation of the voltages and currents in circuits consisting of current and voltage sources, resistors, inductors and capacitors. The basic components, different calculation methods for direct current (DC), first- and second-order circuits are introduced in the first part, EE1C11. In the second part, EE1C21, the techniques presented in the first part are applied to circuits operating in alternating current (AC), while touching upon frequency analysis (filters) and magnetically coupled circuits. The course is wrapped-up by transient analysis via Laplace transform and introducing the two-port concepts. An important part of the first part of Linear Circuits (EE1C11) concerns hands-on experiments in the lab. Here the student learns to read a circuit diagram, to build and engineer an electrical circuit, and to test an electric circuit dealing with real measurements.	
Study Goals	After following this course the student should know: Theoretical / applicative skills: The basic concepts of electrical circuits (current, voltage, charge, energy, power); The basic components as independent sources, resistors, inductors, capacitors, opamps and dependent sources; The laws of Ohm's, Kirchhoff's theorem, Norton and Thevenin equivalents, to calculate currents and voltages in circuits; Serial and parallel circuits to calculate currents and voltages; The nodal method, the mesh method to calculate currents and voltages in circuits; Special instruments for analysing circuits; Methods for analysing first-order transients in electric circuits; Methods for analysing second-order transients in electric circuits. Hands-on skills: How to build circuits making use of soldering techniques; How to read and build a circuit given the diagram of an electrical circuit; To interpret data of components in the data sheets; How to test an electrical circuit making use of a signal-generator, a meter and / or an oscilloscope.	
Education Method	12 hrs lectures 24 hrs instructions 32 hrs lab assignments (course labs) 7 (self-assessed) graded homework assignments handled via the Brightspace platform 2 partial exams	
Literature and Study Materials	Textbook, indicated under "Books". Handouts (copies of the lecture slides) are made available after the lectures the lectures are made available. The answers of the problems done during the instructions are available in the book. Supplementary reading offered via Brightspace.	
Books	Fundamentals of Electric Circuits, 7th edition Charles K. Alexander, Matthew N. O. Sadiku McGraw-Hill ISBN 978-1-260-57079-3	
Assessment	This course has: A course lab, 7 (self-assessed) graded homework assignments handled via the Brightspace platform 2 examinations, (weeks 1.5 and 1.10) A resit. Judgement: The result of the online tests is expressed as R, ranging 0 to 10. The R figure is the average of the results of all the on-line tests, in which a non-made equal to turn into a 0, and wherein any non-made or least well-scored test is omitted. The results of both partial exams are averaged, this gives the exam grade T. The final grade C is calculated as follows: $C = T + R \times (10 - T) / 100.$ This calculation is done with one decimal accuracy. To pass the course, a final grade of at least 5.8 points in scale 10 is needed. The final grade will be rounded to half. The final grade of the course is only valid when the course lab has been successfully passed. The result of the on-line tests, R, is valid for the partial exams and the resit, but only in the year the on-line tests have been made. The resit includes the contents of both partial exams. The result of the partial exams expires for the resit and is not transferable to the next academic year.	
Exam Hours	Partial exams: 2 hrs Resit: 3 hrs	
Permitted Materials during Tests	During the exam, students can use a simple non-programmable and non-graphical calculator (like the TI-30 or Casio FX-82). Students are allowed to have a handwritten A4 with their own summary / notes during the exam. The exam is NOT an open book exam. During the exam, students are allowed to ask questions to clarify the wording.	
Co-Instructor	Dr. I. Ercan	

EE1C21	Linear Circuits B	5
Responsible Instructor	Dr.ing. I.E. Lager	
Instructor	Ir. E.W. Bol	
Instructor	Dr. M. Ghaffarian Niasar	
Instructor	B. Abdikivanani	
Instructor	Dr. F. Fioranelli	
Instructor	Dr. A. Lekic	
Instructor	Dr. I. Ercan	
Contact Hours / Week x/x/x/x	0/6/0/0 hc	
Education Period	2	
Start Education	2	
Exam Period	2 5	
Course Language	English	
Required for	N.A.	
Expected prior knowledge	EE1C11, Linear Circuits A	
Course Contents	This course concerns the SYSTEM analysis of complex, linear circuits. At its core is the calculation of the voltages and currents in circuits consisting of current and voltage sources, resistors, inductors and capacitors. The basic components, different calculation methods for direct current (DC), first- and second-order circuits are introduced in the first part, EE1C11. In the second part, EE1C21, the techniques presented in the first part are applied to circuits operating in alternating current (AC), while touching upon frequency analysis (filters) and magnetically coupled circuits. The course is wrapped-up by transient analysis via Laplace transform and introducing the two-port concepts. An important part of the first part of Linear Circuits (EE1C11) concerns hands-on experiments in the lab. Here the student learns to read a circuit diagram, to build and engineer an electrical circuit, and to test an electric circuit dealing with real measurements.	
Study Goals	After following this course the student should know: - Calculation of voltages, currents and power in circuits with sinusoidal signals using phasors and vector diagrams; - Calculation of instantaneous power, average power, maximum average power, and effective value; - Magnetically coupled circuits, the coupling factor, and the ideal transformer in the analysis of electrical circuits. - Frequency responses, transfer functions, poles, zeros, dB, Bode diagrams and sketches to calculate circuits; - Resonant circuits, filter networks, low pass, high pass, band-pass and band-stop filters, passive and active filters; - The Laplace transform and inverse Laplace, the initial value theorem and the steady-state theorem; - How to apply the Laplace transform to circuits, calculate the transfer function $H(s)$, determine the relationships with poles and zeros, and relations with the frequency response and Bode diagrams; - How to calculate the Y- and Z-parameters of two ports networks and the equivalent parameters of a (serial or parallel) concatenation of two ports.	
Education Method	18 hrs lectures 36 hrs instructions 7 (self-assessed) graded homework assignments handled via the Brightspace platform 2 partial exams	
Literature and Study Materials	Textbook, indicated under "Books". Handouts (copies of the lecture slides) are made available after the lectures the lectures are made available. The answers of the problems done during the instructions are available in the book. Supplementary reading offered via Brightspace.	
Books	Fundamentals of Electric Circuits, 7th edition Charles K. Alexander, Matthew N. O. Sadiku McGraw-Hill ISBN 978-1-260-57079-3	
Assessment	This course has: 7 (self-assessed) graded homework assignments handled via the Brightspace platform 2 examinations, (weeks 2.5 and 2.10) - A resit. Judgement The result of the online tests is expressed as R, ranging 0 to 10. The R figure is the average of the results of all the on-line tests, in which a non-made equal to turn into a 0, and wherein any non-made or least well-scored test is omitted. The results of both partial exams are averaged, this gives the exam grade T. The final grade C is calculated as follows: $C = T + R \times (10 - T) / 100.$ This calculation is done with one decimal accuracy. To pass the course, a final grade of at least 5.8 points in scale 10 is needed. The final grade will be rounded to half. The final grade of the course is only valid when the course lab has been successfully passed. The result of the on-line tests, R, is valid for the partial exams and the resit, but only in the year the on-line tests have been made. The resit includes the contents of both partial exams. The result of the partial exams expires for the resit and is not transferable to the next academic year.	
Exam Hours	Partial exams: 2 hrs Resit: 3 hrs	
Permitted Materials during Tests	During the exam, students can use a simple non-programmable and non-graphical calculator (like the TI-30 or Casio FX-82). Students are allowed to have a handwritten A4 with their own summary / notes during the exam. The exam is NOT an open book exam. During the exam, students are allowed to ask questions to clarify the wording.	
Co-Instructor	Dr. I. Ercan	

EE1C31	Amplifiers and instrumentation	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Instructor	T.M. Lopes Marta da Costa	
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/4/0 pr	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	EE1L21 (EPO-2), EE2L11 (EPO-3), EE2C11 (Integrated Circuits), EE3C11 (Electronics)	
Expected prior knowledge	EE1C11 (Linear Circuits A), EE1C21 (Linear Circuits B), EE1L11 (EPO-1), EE1P11 (Classical and Quantum Physics)	
Course Contents	This course introduces the basics of electronic circuits for processing and amplification of information-carrying signals, and the basics of electronic instrumentation: the acquisition of (non-electrical) information, the conversion of this information to the electrical domain by means of sensors, and the processing and digitization of this information. The course adopts a top-down approach to amplifier design. First, the processing of signals using nullor-based circuits is introduced, which provides an implementation-independent approach to the design of amplifier circuits. Then, implementations of the nullor are introduced, based on the operational amplifier (opamp) and based on single-transistor circuits using bipolar transistors and field-effect transistors. The course covers the analysis of measurement problems and the specification of measurement systems. Various common sources of measurement errors and the concept of uncertainty in measurement results are introduced. Several important instruments for electrical measurements are discussed. Moreover, a number of commonly-used sensors for the measurement of non-electronic quantities are introduced, as well as electronic circuits for the readout of these sensors, including opamp-based readout circuits, instrumentation amplifiers, and analog-to-digital converters.	
Study Goals	After completing this course, the student is able - to describe the transfer function and accuracy of a measurement system - to identify common sources of error in measurement systems and quantify their effect - to describe the operating principles of commonly-used sensors - to analyse circuits with diodes, bipolar transistors and field-effect transistors using small-signal analysis - to synthesize simple circuits for processing of voltages and current signals - to analyse amplifier circuits using two-port models - to synthesize amplifiers based on the nullor two-port - to implement amplifiers using single-transistor circuits - to implement amplifiers using opamps - to apply opamp-based signal-processing circuits for sensor read-out - to quantify errors associated with non-idealities of practical opamps - to describe the operating principles of commonly-used analog-to-digital converters - to describe the operating principle of commonly-used instruments for electrical measurements - to compare available instruments based on quality and accuracy	
Education Method	Lectures, tutorials and labs	
Computer Use	Circuit simulation (LTSpice, Falstad)	
Literature and Study Materials	- Regtien: P.P.L. Regtien, "Electronic Instrumentation," - Reader: J. Hoekstra, "Versterkerschakelingen," - Bell: S. Bell, "A Beginner's Guide to Uncertainty of Measurement," Measurement Good Practice Guide No. 11, Issue 2, National Physical Laboratory. See Brightspace - Course lab manuals: see Brightspace See Brightspace for a detailed list.	
Books	P.P.L. Regtien, "Electronic Instrumentation," second edition, VSSD, 2005. See http://www.vssd.nl/hlf/e008.htm	
Reader	J. Hoekstra, "Versterkerschakelingen," available via https://www.webedu.nl/bestellen/tudelft (can be ordered both as a hardcopy or a (free) pdf; for the exam, you can only use the hardcopy	
Assessment	This course has 6 on-line tests and a final written exam. The result of the on-line tests is expressed as a grade R on a scale from 0 to 10. This grade R is the average of the scores of the individual on-line tests, with the lowest score omitted. Failure to take a test leads to a score of 0 for that test. The result of the final exam is expressed as a grade T on a scale from 1 to 10. The final grade C is calculated as follows: $C = T + R \times (10 - T) / 100.$ The result R of the on-line tests is valid for both exams (at the end of the quarter and the re-sit), but only in the year in which the tests have been taken. Successful completion of the course labs is required to pass the course. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	During the exam, you are allowed to use the reader (Hoekstra), the book (Regtien), a calculator, and a self-made, hand-written, double-sided 1 A4 formula sheet. No other written or printed material is allowed. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	T.M. Lopes Marta da Costa	

EE1D11	Digital Systems A	5
Responsible Instructor	Dr.ir. A.J. van Genderen	
Instructor	Ing. A.M.J. Slats	
Instructor	Dr.ir. J.S.S.M. Wong	
Instructor	M. Taouil	
Contact Hours / Week x/x/x/x	4/0/0/0 hc; 6/0/0/0 pr	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	EE2L11 EPO-3: Design a chip	
Expected prior knowledge	n.a.	
Course Contents	In this course, you will learn as student the basics of digital systems and programming in a procedural programming language, namely C. With respect to the former, this means that you will learn to design, simulate, build and emulate simple digital circuits. With respect to the latter, you will learn to write, compile and execute C programs. To be able to accomplish the above, you will get familiar with several hardware related topics including, among other things, the overall design of computer systems, data representation in computers, boolean algebra, logical minimization, sequential circuits and Finite State Machines. In addition, you will also get an introduction to the hardware description language VHDL. The software related topics include compilation and execution of programs, basic programming constructions (data types, award, choice, and repetition commands), procedural abstraction (methods and parameters) and data abstraction (arrays, structures), use of dynamic data structures such as lists and binary trees, and reading and writing files.	
Study Goals	You should be able to (hardware related part): - explain the basic idea of how digital systems work and how they are build. - describe digital circuits by means of Boolean algebra and able to simplify these circuits (logical minimization). - describe how sequential circuits work, and implement them using Finite-State Machines. - design and analyze simple combinatorial and sequential circuits. - create, simulate and emulate structural VHDL descriptions. You should be able to (software related part): - explain the basic idea of programming, compiling and executing code on a computer. - explain the difference between primitive and composite data types, as well as how they are represented in the computer. - write applications adhering to the syntax of a programming language. - write code that contains control statements such as if-else statements and for/while loops - work with pointers, arrays and structures and use these concepts in dynamic structures such as linked lists and binary trees. - explain how functions and their arguments work, and how to implement iterative and recursive functions. - read from and write to files. - design and implement a simple algorithm.	
Education Method	lectures, hands-on lectures (constructions), lab and practical assignments.	
Computer Use	QuestaSim, Quartus, espresso, https://www.onlinegdb.com	
Literature and Study Materials	- Lecture slides - Lab manual - Digital Design and Computer Architecture, 2nd Edition, David Harris Sarah Harris, Morgan Kaufmann, 2012, ISBN 9780123944245. Optionally: - A Book on C, fourth edition, Al Kelley, Ira Pohl, Pearson Education, ISBN 0-201-18399-4. - The C Programming Language, 2nd edition, B.W. Kernighan, D.M. Ritchie, Prentice Hall Software Series, ISBN 9780131103627. - Student's Guide to VHDL, P.J. Ashenden, Morgan-Kaufmann, 2nd edition 2008, ISBN 978-155-860865-8.	
Assessment	Practical assignments, lab, on-line tests, 2 partial exams, and a re-exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Judgement: The result of the online tests is expressed as R, ranging 0 to 10. The R figure is the average of the results of all the on-line tests, in which a non-made test results into a 0. The results of the two partial exams are averaged. This gives the exam grade T. The final grade of C of the course is then calculated as follows: $C = T + R \times (10-T) / 100.$ This calculation is done with one decimal accuracy, and the final grade will be rounded up to half. The result of the on-line tests, R, is valid for the partial exams and the resit, but only in the year the on-line tests have been made. The final mark is valid only when all course labs and practical assignments have been successfully completed. The resit includes the contents of both partial exams. The result of the partial exams expires for the resit and is not transferable to the next academic year.	
Permitted Materials during Tests	Study material, including lecture slides	
Co-Instructor	Dr.ir. J.S.S.M. Wong	
Co-Instructor	M. Taouil	

EE1D21	Digital Systems B	5
Responsible Instructor	Dr.ir. A.J. van Genderen	
Instructor	Ing. A.M.J. Slats	
Instructor	Dr.ir. J.S.S.M. Wong	
Contact Hours / Week x/x/x/x	4/0/0/0 hc	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Expected prior knowledge	EE1D11 Digital Systems A	
Course Contents	The course Digital Systems B is a follow-up to the course Digital Systems A and continues on the hardware of digital systems, and the hardware description language VHDL. Topics that are addressed include: implementation technology (including FPGAs), standard modules, RTL systems and data path control model, a simple processor (MIPS) and assembly language, VHDL behavior and configuration descriptions, synthesis methods and VHDL tools.	
Study Goals	- Have knowledge about modern implementation technologies for digital circuits - Have knowledge about combinatorial and sequential modules used in modern digital circuits, and how these modules can be used during design. - Have knowledge about the structure of a processor and how to program a processor in assembly language. - Have knowledge about serial communication protocols such as I2C, SPI and UART. - Conceptually recognize the hardware software design space available for designing digital applications. - Design a simple application entirely in hardware, for example, if high performance is paramount. - Design the same application (partly) in the form of software that is running on already designed hardware (eg a standard microprocessor), for example, if flexibility is paramount. - Optimize (synthesize) a behavioral specification to a logic circuit at gate-level (eg with FPGAs) using software synthesis tools.	
Education Method	lectures, hands-on lectures and practical assignments.	
Literature and Study Materials	- Digital Design and Computer Architecture, 2nd Edition, David Harris Sarah Harris, Morgan Kaufmann, 2012, ISBN 9780123944245 - Optionally: Student's Guide to VHDL, P.J. Ashenden, Morgan-Kaufmann, 2nd edition 2008, ISBN 978-155-860865-8. - Lecture slides.	
Assessment	Practical assignments, on-line tests and written exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Judgement: The result of the online tests is expressed as R, ranging 0 to 10. The R figure is the average of the results of all the on-line tests, in which a non-made test results into a 0. Let the exam grade be given by T. The final grade C for the course is then calculated as follows: $C = T + R \times (10 - T) / 100.$ This calculation is done with one decimal accuracy, and the final grade will be rounded up to half. The result of the on-line tests, R, is valid for the initial exam and the resit, but only in the year the on-line tests have been made. The final mark is valid only when all practical assignments have been successfully completed.	
Permitted Materials during Tests	Study material, including lecture slides	
Co-Instructor	Dr.ir. J.S.S.M. Wong	

EE1L11	EPO-1: Booming Bass	5
Responsible Instructor	Dr.ir. N.P. van der Meijs	
Instructor	Dr.ing. I.E. Lager	
Instructor	Ir. E.W. Bol	
Instructor	Dr.ir. M. Alonso Del Pino	
Instructor	S. Izadkhast	
Instructor	B. Abdikivanani	
Instructor	Dr. A. Lekic	
Instructor	Dr. I. Ercan	
Instructor	K.M. Dowling	
Contact Hours / Week x/x/x/x	0/8/0/0 pr	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Build, analyze and characterize a sound system.	
Course Contents Continuation	In the EPO-1 project you and your team will design a complete audio system with "Booming Bass" extension. First, you will measure the characteristics of a 3-way loudspeaker system. You will analyze and build a power supply and a power amplifier. You will design speaker filters and the extension for a so-called "Booming Bass." Starting from a set of specifications, you will make design choices for each part of the system. Within your team you will discuss the choices and divide the work amongst each other. To verify the choices you will make analytical calculations and numerical simulations. For this purpose you will use the mathematics and the linear circuits theory from previous and concurrent lecture courses. You will construct the designs and test them with electrical and acoustic equipment. The measurement results are compared with the analytical and simulation results. On this basis, you will conclude to which extent the specifications have been met. During EPO-1 you will also train team working skills. Those include working in organized meetings, making a project plan, giving oral presentations, and delivering technical reports. Your project results will be graded on the basis of the reports, an oral presentation, and an individual oral examination. The sound of the system will be assessed in an informal competition where all students are welcome. A jury of experts will select a winning team and award a small prize.	
Study Goals	After completion of this project the student should master the following: 1) The integration in circuit analysis and design of important concepts of the theoretical lecture courses of the first semester. This includes: - Application of resistors, capacitors, inductors, and operational amplifiers. - Determination of transfer functions of simple circuits with those components in the time domain as well as in the frequency domain. - Simulation of the circuits in Matlab and PSpice. - Design of 1st and 2nd order filters. - Design of a simple opamp voltage amplifier. - Analysis of a power supply with transformer and diode bridge. 2) Basic skills of electronic experimentation: - Soldering of components on a printed circuit board with the aid of a schematic. - Effective use of a multimeter, function generator and oscilloscope. - Interpretation of measurement results in the light of non-idealities of measurement equipment and components. 3) The basics of working in a project team. This includes: - The planning of work, - Organizing structured meetings, - Writing of a project plan, - Oral presentation, and - Writing of engineering reports. 4) Academic skills on a basic level concerning: - Quickly accessing a specific problem, - Retrieving of technical information in data sheets and manuals, - Critical assessment and citation of information. 5) A good study attitude: - Motivation for the Electrical Engineering degree course, - Working sufficiently hard and regularly, - Delivering in time, - Working sufficiently independently.	
Education Method	Practical: group of about 10 students with a tutor and a student assistant. Contact hours (Online and on-campus): 48 hours (6 hours/week) on average. Self study: 92 hours	
Practical Guide	Student manual of Course Labs of EE1C11 the EPO-1 Project EE1L11, 2020-2021. For sale at the ETV.	
Books	Roel Grit, Projectmanagement, 6th ed., Noordhoff Uitgevers B.V. (For sale at the ETV.) Elling et.al., Report Writing for Readers with little Time. Noordhoff Uitgevers B.V., 2012. (For sale at the ETV https://etv.itdepartment.nl/).	
Prerequisites	Completion of EE1C11 Course labs of Q1	
Assessment	The EPO-1 project will be graded on the basis of the following components: 1. Final report. Group grade; it will constitute 25% of the total grade. 2. Intermediate report. Group grade; it will constitute 15% of the total grade. 3. Individual oral examination in Week 2.9. Individual grade, constituting 30% of the total grade.	

4. Oral presentation skills in Week 2.9. Individual grade, constituting 15% of the total grade.
5. Tutor individual grade taking into account your overall performance in the project 15%;
6. Individual tasks during the project. Individual grade, 'Pass' or 'Fail'. During the project all members should have brought to a good end at least the following tasks:
 - Giving a presentation on the content of an intermediate report.
 - Answering questions on the content of an intermediate report.
7. Mentoring assignment. Individual grade, 'Pass' or 'Fail'.

Required for obtaining a final grade for this course is that all the components individually have scored a 'Pass' or a grade of at least

Grading during the project will be done by the tutor. The examinations at the end will be graded by a panel consisting of two or three people who are members of the academic staff of the EE program.

Assessment will be done on the basis of the study goals. You are advised to consult them from time to time in order to structure your study activities.

The deadlines in this project are hard, especially the one for the final report. Work that is handed in too late may still be graded. However, it will be subject to a penalty in such a way that the maximum obtainable grade will be a 6 ('sufficient').

Components that at the first assessment are judged as insufficient, will not be graded. To correct this, one chance will be given, after which a grade will be determined.

EPO-1 includes the mentorship. The grade of the EPO-1 project becomes not valid until all of the following mentorship requirements have been fulfilled:

- presence and active participation in the mentorship activities including Goal Setting tasks during the 1st semester; ITAV Teams work and other activities.

Tags

Circuits
 Design
 Elektriciry & Magnetism
 Group work
 Lab Research
 Matlab
 Modelling
 Practicals
 Project planning / management
 Small groups

Co-Instructor

Dr.ing. I.E. Lager

Co-Instructor

S. Izadkhast

EE1L21	EPO-2: Smart Robot Challenge	5
Responsible Instructor	M. Spirito	
Instructor	Dr.ir. A.J. van Genderen	
Instructor	Ing. R.M.A. van Puffelen	
Instructor	Ir. E.W. Bol	
Instructor	S. Izadkhast	
Contact Hours / Week x/x/x/x	0/0/0/8 pr	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	EE1C31, EE1D11, EE1D21	
Course Contents	Build, program and operate a functional autonomous mobile robot. The students will have to apply the acquired knowledge and skills in the area of digital control electronics, sensor technology and design of analog interface electronics.	
Course Contents Continuation	The EPO2 project 'Smart Robot Challenge' is dedicated to design and build a wireless controlled robot that can autonomously perform a predefined set of tasks. The basic hardware is available at the beginning of the course, i.e., robot platform, Zigbee modules, battery and optical sensors. The hardware must be extended where necessary to allow the robot to sense and identify various obstacles placed in the tasks to be performed. C programming and VHDL language are employed so that the robot can complete the various challenges and win against other teams robot the final competition.	
Study Goals	Upon successful completion of EPO2 project, the student will be able to: 1. Apply a project-based approach to design an electronic system in a project team; 2. Methodologically design a technical solution to a relevant problem within given boundary conditions; 3. Integrate and apply the knowledge and skills acquired during the lecture classes of the first year, in a design project; 4. Actively apply academic skills in order to intenfity and acquire missing information and knowledge, systematically analyze problems, motivate choices, critically interpret results; 5. Document own project-related work and write a well-organized technical report.	
Education Method	Practical: group of 8-10 students with 1 tutor and 2 student assistants. Contact hours: 64 hours (8 hours/week) Self study: 64 hours (8 hours/week + preparation for Final presentation)	
Literature and Study Materials	The practicum manual is available on the course EE1L21 Brightspace.	
Practical Guide	Student manual EPO-2 project: "Smart Robot Challenge"	
Prerequisites	Completion of course labs of both EE1D21 and EE1C31 in Q3	
Assessment	1) Final project report 2) Oral group presentation 3) Robot tests/competition	
Remarks	EPO-2 includes the mentorship. The grade of the EPO-2 project becomes not valid until all of the mentorship requirements are completed. This will be registrated with a pass/fail The result of this project, both on group- and individual level, requires a large commitment of all students. In case of a lack of commitment or insufficient preparing (self study) further participation in the project can be denied.	
Judgement	The final grade will be calculated from: Final report	

EE1M11	Linear Algebra and Analysis A	5
Responsible Instructor	Dr. J.W. van der Woude	
Contact Hours / Week x/x/x/x	6/0/0/0 hc	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	High school mathematics (Wiskunde B)	
Course Contents	In this course basic concepts from Linear Algebra and Analysis are treated. The course starts complex numbers, followed by methods from Linear Algebra to solve and analyse systems of linear equations. Here concepts as vectors, linear combination, span and linear independence play a central role. Also matrices and matrix algebra will be discussed. Attention is paid to the vector representation of lines and planes in space, the inner product (dot product), the cross product and orthogonality. Next, subjects from Analysis are treated, to begin with linear differential equations of first and second order with constant coefficients. Then, standard functions from high school are repeated, followed by an introduction of their inverse functions. Subsequently, limits and the squeeze theorem are treated, followed the rule of l'Hospital, implicit differentiation, linearisation, differentials and Taylor polynomials. Finally, Riemann sums, the fundamental theorem of calculus, the primitive, the substitution rule, integration by parts, partial fraction decomposition and improper integrals will be treated.	
Study Goals	Students should be able to recognise and apply the topics mentioned in the Course Contents within their Electrical Engineering program.	
Education Method	The course consists of weekly 3 x 2 hours colstruction (a combination of lecture and instruction) in groups of around 50 students. In each colstruction the theory will be explained shortly, whereupon the students have ample time to work under guidance on the exercises. On the Brightspace page of the course pre-lecture videos, extra material and presentations can be found by which the students can prepare themselves for the colstruction. On the Brightspace page also the exercises per colstruction can be found, and a link to Grasple, a digital platform, by which the students can practice with the material afterwards. Grasple is also the platform used for the progress tests.	
Literature and Study Materials	The Lineair Algebra part is based on: David Poole, "Linear Algebra (a modern introduction)", Brooks/Cole, ISBN-13: 9781285463247, (4e edition). The Analysis part is based on: "Calculus: Early Transcendentals", by James Stewart, edition 8, ISBN: 9781305272378. Also can be used: "Calculus: Early Transcendentals, Metric Edition", by James Stewart, Daniel Clegg and Saleem Watson, edition 9, ISBN: 9780357113516. Conversion tables are available.	
Assessment	Calculator Only a non-programmable calculator without capabilities for storage of text or formulas and without capabilities for graphing may be used. Suitable types are the TI-30 and the Casio FX-82, and the various variants of these machines. An entrance test (in week 1.1, formative, as a part of the first year mentoring). A trial-exam (in week 1.4, formative, as a part of the first year mentoring). A sub-exam (nl: deeltentamen) on the first half of the course contents in week 1.5. A sub-exam on the second half of the course contents in week 1.10. A resit (nl: herkansing) over the entire course contents (i.e., including all topics) in week 2.7 The course also contains four progress tests for which participation is voluntary and by which a little bonus can be earned. During the sub-exams and resit only a non-programmable calculator without capabilities for storage of text or formulas and without capabilities for graphing may be used. Suitable types are the TI-30 and the Casio FX-82, and the various variants of these machines. Not allowed is, for instance, the Casio FX-991EX. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The result of the progress tests is expressed in a grade R on the scale from 0 to 10. This grade R is the average of the results of all progress tests, where a test not made returns a result equal to 0, and where one test with the lowest score is omitted in computing the average. The result of the two sub-exams will be averaged, this will give the grade T. The final grade C of the course is calculated as follows: $C = T + R \times (10 - T) / 100$. The calculation takes place by one decimal place, and the final grade is rounded off to the nearest integer or half. The result R of the progress tests is valid for both examinations of the course (the first final exam and the resit), but only in the year that the progress tests are made. The resit is taken in one part (i.e., contains all course topics). The result of the sub-exams expires before the resit, and is not transferable to the next academic year.	
Co-Instructor	Dr. B.J. Meulenbroek	
Co-Instructor	N. de Kleijn	
Co-Instructor	Dr. N.D. Verhulst	
Co-Instructor	Dr. H.A.W.M. Kneppers	

EE1M21	Linear Algebra and Analysis B	5
Responsible Instructor	Dr. J.W. van der Woude	
Contact Hours / Week x/x/x/x	0/6/0/0 hc	
Education Period	2	
Start Education	2	
Exam Period	2 5	
Course Language	English	
Expected prior knowledge	EE1M11	
Course Contents	Functions of several variables, partial derivatives, tangent planes and linear approximations, the chain rule, directional derivatives and the gradient vector, double integrals over rectangles, double integrals over general regions, double integrals in polar coordinates, applications of double integrals, triple integrals, triple integrals in cylindrical coordinates, triple integrals in spherical coordinates, change of variables in multiple integrals, vector fields, line integrals and arc length of a curve, the fundamental theorem for line integrals, Green's theorem, curl and divergence, parametric surfaces and their areas, surface integrals, Stokes' theorem, the divergence theorem.	
Study Goals	Students should be able to recognise and apply the topics mentioned in the Course Contents within their Electrical Engineering program.	
Education Method	The course consists of weekly 3 x 2 hours colstruction (a combination of lecture and instruction) in groups of around 50 students. In each colstruction the theory will be explained shortly, whereupon the students have ample time to work under guidance on the exercises. On the Brightspace page of the course pre-lecture videos, extra material and presentations can be found by which the students can prepare themselves for the colstruction. On the Brightspace page also the exercises per colstruction can be found, and a link to Grasple, a digital platform, by which the students can practice with the material afterwards. Grasple is also the platform used for the progress tests.	
Books	The course is based on: "Calculus: Early Transcendentals", by James Stewart, edition 8, ISBN: 9781305272378. Also can be used: "Calculus: Early Transcendentals, Metric Edition", by James Stewart, Daniel Clegg and Saleem Watson, edition 9, ISBN: 9780357113516. Conversion tables are available.	
Assessment	A sub-exam (nl: deeltentamen) on the first half of the course contents in week 2.5. A sub-exam on the second half of the course contents in week 2.10. A resit (nl: herkansing) over the entire course contents (i.e., including all topics) in quarter 5, week 5.3/5.4. The course also contains four progress tests for which participation is voluntary and by which a little bonus can be earned. During the sub-exams and resit only a non-programmable calculator without capabilities for storage of text or formulas and without capabilities for graphing may be used. Suitable types are the TI-30 and the Casio FX-82, and the various variants of these machines. Not allowed is, for instance, the Casio FX-991EX. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The result of the progress tests is expressed in a grade R on the scale from 0 to 10. This grade R is the average of the results of all progress tests, where a test not made returns a result equal to 0, and where one test with the lowest score is omitted in computing the average. The result of the two sub-exams will be averaged, this will give the grade T. The final grade C of the course is calculated as follows: $C = T + R \times (10 - T) / 100$. The calculation takes place by one decimal place, and the final grade is rounded off to the nearest integer or half. The result R of the progress tests is valid for both examinations of the course (the first final exam and the resit), but only in the year that the progress tests are made. The resit is taken in one part (i.e., contains all course topics). The result of the sub-exams expires before the resit, and is not transferable to the next academic year.	
Co-Instructor	Dr. B.J. Meulenbroek	
Co-Instructor	N. de Kleijn	
Co-Instructor	Dr. N.D. Verhulst	
Co-Instructor	Dr. H.A.W.M. Kneppers	

EE1M31	Probability and Statistics	5
Responsible Instructor	Dr. C.O. Smet	
Contact Hours / Week x/x/x/x	0/0/0/2 hc; 0/0/0/4 wc	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Calculus: integration and differentiation in one and two variables. Linear Algebra: row reduction, matrix multiplication, normal equations	
Course Contents	Graphical and numerical summaries of data Introduction to probability Some elementary combinatorics to find probabilities Conditional probability; Bayes' rule; stochastic (in)dependence Random variables; probability mass function; density function; distribution function Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto Expectation and variance; transformations; Jensen's inequality Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables Covariance and correlation Chebychev's inequality; Law of Large numbers; Central Limit Theorem Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot Theory of Estimators: Bias, Efficiency, Mean squared error Linear Regression: univariate and multivariate, goodness of fit Introduction to statistical learning Confidence intervals of Mean and Proportion; t-distribution Testing theory: Type I/II Error, p-value, significance level, critical region One sample t-test Parametric bootstrap	
Study Goals	At the end of the course, the students will be able to: - compute (conditional) probabilities - use discrete and continuous random variables - use joint random variables - compute expectation and (co)variance - apply the law of large numbers and the central limit theorem - analyse data by means of graphical and numerical summaries - formulate a statistical model - estimate parameters - perform a linear regression analysis - construct a confidence interval - formulate and carry out hypothesis tests - perform a bootstrap analysis - Use Python to perform elementary simulation and data analysis	
Education Method	Twice per week "Colstruction" in blended form: the student watches a prelecture, makes exercises related to the video, comes to class, participates in the lecture (combination of lecturing and making exercises) and works at additional exercises at home. Once a week there is "Instruction": the student can ask question about the exercises and work on exam-like exercises.	
Books	A Modern Introduction to Probability and Statistics Understanding Why and How Series: Springer Texts in Statistics Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E. 2005, XVI, 488 p. 120 illus., Hardcover ISBN: 978-1-85233-896-1	
Assessment	One midterm (week 5) on Part 1: Probability Theory and one final exam (week 10) on Part 2: Statistics. Final grade is the average of both tests. A maximum of 1 bonus point can be earned for the computer assignments provided that the final grade is greater than or equal to 5.0. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Formula sheet (will be provided at each test) and non-graphical, non-programmable calculator.	
Tags	Stochastics	
Judgement		

EE1P11	Classical and Quantum Mechanics	5
Responsible Instructor	M. Mastrangeli	
Instructor	Dr.ir. R. Santbergen	
Instructor	Dr. I. Ercan	
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/4/0 wc	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Expected prior knowledge	High school-level physics, including vectors; calculus, including complex numbers and differential equations.	
Course Contents	EE1P11 "Classical and Quantum Mechanics" teaches the basic elements of physics that are required to become an electrical engineer. In the first part of this course the theory of classical mechanics is constructed based on observable phenomena, most of which will be demonstrated experimentally. It will be shown that motion (velocity, acceleration) is caused by forces that can be described by vectors in three dimensions. When forces act on particles or objects, they cause translation, rotation or wave motion. It will be shown that forces can also be used to introduce concepts like work and energy. These concepts can in turn be used fruitfully to describe complex motion caused by time- and position-dependent forces, by applying laws of conservation of kinetic or potential energy. In the second part of this course the non-intuitive physics ruling the behaviour of matter and its interaction with radiation at atomic scale, i.e. quantum mechanics, will be introduced. It will be shown that particles have wave-like behaviour and vice versa, and that on atomic scale the energy of particles varies in discrete steps rather than continuously as is the case in classical mechanics. Heisenbergs uncertainty principle will also be discussed. In this part students will learn to carry out calculations on standard quantum mechanical examples like the particle-in-a-box using the Schrödinger equation. Finally, band diagrams and the conduction of current in metals and semiconductors will be described, as well as a qualitative description of the working principle of the diode, the transistor and the solar cell.	
Study Goals	After successful completion of this course, the student should be able to: For Classical Mechanics: - Understand and explain the fundamental laws of classical mechanics, illustrate examples, and apply these to practical situations. - Translate observation into mathematical equations using general laws and subsequently express unknown quantities into known quantities. - Carry out calculations on the movement and rotation of objects that experience force or torque in three dimensions by making use of the vector quantities, like angle and position, (angular) velocity, and (angular) acceleration. - Describe the harmonic oscillator and apply this concept to several harmonic oscillator systems. - Apply the knowledge of wave phenomena like wave propagation and interference using trigonometric equations. For Quantum Mechanics: - Realize the need for non-classical physics to account for the experimental behavior of matter and radiation at atomic scale. - Be familiar with the physical behavior of particles like electrons and photons, and to determine the appropriate energies and wavelengths in several situations. - Realize how Heisenbergs uncertainty principle limits the accuracy in simultaneous knowledge of some observables of particles. - Carry out simple calculations using the Schroedinger equation. - Understand the band structure of a solid and apply this concept to describe the working principle of electronic devices such as the diode, the transistors and the solar cell.	
Education Method	From week 1 until week 4 there will be 2 sessions of 2-hour lectures and 2 sessions of 2-hour instructions in Classical Mechanics. Likewise, from week 6 until week 9 there will be 2 sessions of 2-hour lectures and 2 sessions of 2-hour instructions in Quantum Mechanics	
Books	The following book will be used for this course: Richard Wolfson, "Essential University Physics", vol. 1 and 2, 4th Global Edition, Pearson. For the quantum mechanics part of the course, a Reader is also made available from the instructors. In addition the following digital learning environment may be used: Mastering Physics (http://www.pearsonmylabandmastering.com/northamerica/masteringphysics/) This digital learning environment will be used to support the instructions. For this digital learning environment you will need to use your own laptop.	
Assessment	The assessment of this course is based on examinations and progress tests. This is organized as follows: - During the lecturing period of this course, self-graded homeworks (SGHs) will be assigned, whose evaluation contributes to the final grade of the course (see below). - In week 5 a test for the part on Classical Mechanics will be offered. - In week 10 a test for the part on Quantum Mechanics will be offered. - At the resit examination offered in July the entire course (that is, Classical and Quantum Mechanics) will be examined. Judgement: The result of the self-graded homeworks is expressed in a mark, P, on a scale of 0 to 10. P is the average result of all SGHs in which a test that has not been made leads to a 0. For the calculation of P the lowest scoring test is not taken into account. The mark P of the progress tests is valid for both examinations (the first examination through mid-term exams, and the resit), but only in the academic year in which these progress tests have been made. The mark of the examination is expressed with the letter E. This examination mark is the average of the two sub-examinations for the two parts of this course during the first opportunity, or the mark for the examination of the entire course during the second (resit) opportunity. The end result for this course, F, is calculated as follows: $F = E + [P \times (10 - E)/100]$. This calculation is carried out using one decimal and the end result F is rounded to halves. Please note that the impact of P on F is limited, its impact the higher the smaller E. The end result is only valid if all course labs have been finished satisfactorily.	
Judgement	The result of the self-graded homeworks is expressed in a mark, P, on a scale of 0 to 10. P is the average result of all SGHs in which a test that has not been made leads to a 0. For the calculation of P the lowest scoring test is not taken into account. The mark P of the progress tests is valid for both examinations (the first examination and the resit), but only in the academic year in which these progress tests have been made. The mark of the examination is expressed with the letter E. This examination mark is the average of the two sub-examinations for the two parts of this course during the first opportunity, or the mark for the examination of the entire course during the	

second (resit) opportunity.

The end result for this course, F, is calculated as follows: $F = E + [P \times (10 - E)/100]$. This calculation is carried out using one decimal and the end result F is rounded to halves. Please note that the impact of P on F is limited, its impact the higher the smaller E.

The end result is only valid if all course labs have been finished satisfactorily.

Co-Instructor

Dr. I. Ercan

EE1P21	Electricity and Magnetism	5
Responsible Instructor	Dr.ing. I.E. Lager	
Responsible Instructor	Prof.dr.ing. A. Neto	
Contact Hours / Week x/x/x/x	0/0/0/4 hc; 0/0/0/2 wc	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Electric charge, force and field Gauss's law Electric potential Electrostatic energy and capacitors Electric current Magnetism, force and field Electromagnetic induction Maxwell's equations and Electromagnetic waves	
Study Goals	Students have to develop a basic understanding of the theory of electricity and magnetism. Student have to be able to describe the following concepts: electric charge, electric force, electric dipoles, electric field, electric field lines, electric potential, energy stored in the electric field, magnetic force, magnetic dipoles, magnetic field, magnetic flux, magnetic fieldlines, induction, electromagnetic waves Students have to be able to do calculations with the following quantities: electric charge, electric force, electric dipoles, electric field, electric potential, energy stored in the electric field, magnetic force, magnetic dipoles, magnetic field, magnetic flux, induction, electromagnetic waves. In these calculations, students have to be able to use the laws of Coulomb, Gauss, Biot-Savart, Ampere, Faraday, Lenz en Maxwell.	
Education Method	Lectures, instructions, Mastering Physics	
Computer Use	Mastering Physics	
Literature and Study Materials	The course's textbook is Richard Wolfson: Essential University Physics, 4th edition, vol. II, Pearson, 2021 (the textbook, together with an access code for Mastering Physics, are available via the ETV). Lecture handouts, a complete course reader, a solution manual and a substantial amount of supplementary material is provided via the course's Brightspace platform. The course makes use of the Mastering Physics digital learning environment - this instrument is used exclusively for practicing purposes. Note: Students interested in more advanced topics may consider consulting Harald J.W. Müller-Kirsten, Electrodynamics, 2nd ed., World Scientific, 2011 (this book is also available via the ETV).	
Assessment	This course has graded homework assignments and two partial exams. The result of the graded homework assignments is expressed in a mark R between 0 and 10. This mark R is the average of the results of all graded homework assignments, where the result of the graded homework assignment with the lowest mark does not count. The marks of the two partial exams is averaged to the mark for the exam T. The final mark for the course is calculated as $C = T + R \times (10 - T) / 100$. The calculation is done with one decimal, and the final mark C is rounded to halves. The result of the graded homework assignments is valid for both exams (the two partial exams and the resit), but only in the year the homework assignments have been made. The resit is an exam for the whole course, covering both electricity and magnetism. The result of the partial exams is not valid for the resit and is not valid in the next year.	
Permitted Materials during Tests	A form with equations is available. It is a closed book exam. Only not-programmable calculators may be used. Examples of suitable types are the TI-30, the Casio FX-82, and variants of these calculators.	
Co-Instructor	Dr.ing. I.E. Lager	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Electrical Engineering

B-EE jaar 2 2022	
Director of Studies	Dr.ir. N.P. van der Meijs
Program Coordinator	L.M. Ramirez Elizondo
Program Coordinator	H. Shekarian Tabarestani
Prerequisites	Let op! De Ingangseisen zijn in de OER vastgelegd. Mocht er verschil zijn tussen wat er hier staat en wat in de OER staat, dan is datgene wat in de OER staat geldig. In het tweede jaar gelden voor de volgende twee studieonderdelen ingangseisen: Je mag pas aan het onderdeel EE2L11 EPO-3: Ontwerp een Chip meedoen als je de vakken EE1L11 EPO-1: De Beukende Bas, EE1D11 Digitale Systemen A en EE1D21 Digitale Systemen B en het practicum van het vak EE2C11 Geïntegreerde Schakelingen hebt behaald. Je mag pas aan het onderdeel EE2L21 EPO-4: "KITT" zelfsturende auto meedoen als je het vak EE1L21 EPO-2: Smart Robot Challenge en het practicum van het vak EE2T11 Telecommunicatie A hebt behaald.
Introduction 1	Inschrijven Voor de studieonderdelen EE2L11 EPO-3: Ontwerp een Chip en EE2L21 EPO-4: KITT zelfsturende auto moet je je uiterlijk twee weken voor de start van het kwartaal waarin ze plaatsvinden via Brightspace inschrijven. Voor de (deel-)tentamens van alle vakken moet je je uiterlijk 1 week vóór de dag van het tentamen via OSIRIS inschrijven. Wil je bij nader inzien toch niet meedoen aan een studieonderdeel of tentamen waar je je voor hebt ingeschreven? Schrijf je dan ook weer uit! Zie voor uitleg https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens
Introduction 2	Vooruitblik jaar 3: In het derde jaar volg je tijdens het eerste semester een minor. Deze kies je al dit jaar. Kijk voor meer informatie op http://minors.tudelft.nl. Wil je in je derde jaar naar het buitenland? Begin je dan nu al te oriënteren op https://www.tudelft.nl/studenten/ewi-studentenportal/onderwijs/study-abroad Zie ook de Brightspace van Study Abroad EEMCS

EE2C11	Integrated Circuits	5
Responsible Instructor	Dr.ir. N.P. van der Meijs	
Contact Hours / Week x/x/x/x	4/0/0/0 hc; 4/0/0/0 pr	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	The course labs of EE2C11 need to be completed as a pre-requirement for EE2L11	
Expected prior knowledge	Linear Circuits A and B, Amplifiers and Instrumentation, Digital Systems A and B	
Course Contents	This course provides an introduction to analyzing and designing digital integrated circuits, in which both the circuit aspects and the system aspects will be studied. The opportunities and constraints associated with the realization of a digital system on a chip are taken as a starting point. Strong emphasis is placed on working with (approximate) models for physical artifacts. This course prepares for the EE2L11 project, delivering both theoretical knowledge and practical skills.	
Study Goals	Basic concepts of integrated digital circuits and the principles of design and manufacturing. Operation of CMOS transistors, transistor models, inverters and complementary static logic, static and dynamic behavior. Memory elements. RC modeling of interconnect. Design as a trade-off between different requirements such as area and speed. System perspective: timing properties, architecture and realization of larger building blocks for eg arithmetic functions.	
Education Method	Lectures, tutorials, course labs	
Computer Use	Python, SPICE (LTSpice) and other tools for chip design.	
Literature and Study Materials	"Digital Integrated Circuits: A Design Perspective", 2nd edition, J.M. Rabaey et al., Prentice Hall, 2003, ISBN: 0131207644 EE2L11 syllabus via Brightspace	
Practical Guide	EE2L11 lab manual	
Assessment	This course includes course labs (practical exercises) and a final written exam. The final mark for the course is the mark for the exam, but it is only valid when all course labs (practical exercises) have been completed with sufficient results. The course labs expire at the end of the academic year. This means that if you do not have a sufficient grade for the exam at the end of the year, you have to redo the labs during the next year.	
Permitted Materials during Tests	The exam is a closed book exam, but the tables and formulas of the inside of the front flap and rear flap of "Rabaey" will be made available. In addition, a single-sided self-handwritten A4 as a cheat sheet is permitted. Only a non-programmable calculator without capabilities for storage of text or formulas and without capabilities for graphing may be used. Suitable types are the TI-30 and the Casio FX-82, and the various variants of these machines.	

EE2E11	Electrical Energy Conversion	5
Responsible Instructor	Dr.ir. P. Bauer	
Instructor	J. Dong	
Responsible for assignments	Ir. G.J.M. Koeners	
Co-responsible for assignments	H.J.M. Olsthoorn	
Contact Hours / Week x/x/x/x	4/0/0/0 hc; 4/0/0/0 pr	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Principles of electrical power conversion, top down approach system and its components (storage, power electronic and electromechanical conversion), power theory of complex voltage and current waveforms, principles of DC-DC power electronic converters, electrical machines and drives	
Study Goals	After taking this course, students should be able to LO1: Electrical system and energy storage (2 lecture hours, 7.7%) 1.1 Describe the principles of sources and energy storage elements. 1.2 Identify electrical system components; 1.3 Analyse DC and AC circuits. 1.4 Identify the difference between power and energy. 1.5 Calculate the power and energy. LO2: Phasors, power and three-phase (2 lecture hours, 7.7%) 2.1 Explain active and reactive power. 2.2 Identify difference between average and instantaneous power. 2.3 Calculate power factor, distortion factor and total harmonic distortion. 2.4 Calculate complex power in phasors. 2.5 Calculate voltage, current and power in symmetric three-phase systems. LO3: Magnetic circuit, inductor and transformer (3 lecture hours, 12%) 3.1 Apply Amperes law and Faradays law in system of coils. 3.2 Describe the characteristics of magnetic material. 3.3 Describe the construction basics of inductors and transformer. 3.4 Solve simple magnetic circuit with and without air-gap. 3.5 Explain the operation principle of transformer. 3.6 Describe the leakage inductance and how it affects the transformer performance. 3.7 Apply the transformer equivalent circuit and calculate referred impedance. LO4: Mechanical dynamics (1 lecture hour, 3.9%) 4.1 Describe the functioning of inertia, pulley & belt and gearboxes. 4.2 Calculate the steady state operation of system containing mechanical components. LO5: Power electronics, DC-DC converters (8 lecture hour, 30.8%) 5.1 Explain the operation of power electronics switches. 5.2 Explain the operation of pulse width modulation of power electronics converters. 5.3 Describe the energy flow in inductors and capacitors of DC-DC converters. 5.4 Analyse calculate the quasi-static performance of buck, boost, buck-boost, flyback and 2-quarant DC-DC converters. 5.5 Identify the DCM and CCM of DC-DC converters. LO6: DC machine and drives (4 lecture hours, 15.4%) 6.1 Explain the construction and operation principle of DC machines. 6.2 Describe the power flow, calculate efficiency and losses. 6.3 Explain the nameplate of a DC machine. 6.4 Apply induced voltage equation, torque equation and equivalent circuit of DC machines. 6.5 Calculate of the torque, speed and electrical excitation characteristics of separately excited DC machines. 6.6 Explain four quadrant operation and flux weakening of DC drive. LO7: AC machines and drives (6 lecture hours, 23.1%) 7.1 Explain the construction and operation principle of AC machines (synchronous machine and induction machine). 7.2 Explain the rotating magnetic field. 7.3 Explain and calculate the slip of induction machine. 7.4 Apply equivalent circuits of AC machines. 7.5 Apply phasors to calculate synchronous machine performance. 7.6 Explain and calculate the power angle of synchronous machine. 7.7 Describe the power flow, calculate efficiency and losses of AC machines. 7.8 Explain basic structure and principle of AC drive. 7.9 Explain the operation principle of three-phase AC inverter. 7.10 Calculate output voltage of three-phase inverter from modulation index or vice versa.	
Education Method	Lectures and laboratory practicals	
Literature and Study Materials	Reader partially based on the book The Principles of Electronic and Electromechanical Power Conversion A Systems Approach, Braham Ferreira, Wim van der Merwe, 1st edition Wiley, January 2014,	
Assessment	The course is evaluated by a written examination and pass/no-pass of the practicals. The final grade is only valid when all course labs are completed and passed. The course lab marks expire at the end of the academic year. This means that when a student does not pass the corresponding exam and resit in a particular academic year, then the labs must be repeated in the subsequent year.	
Co-Instructor	J. Dong	

EE2E21	Sustainable Energy Supply	5
Responsible Instructor	P. Manganiello	
Responsible Instructor	Dr. M. Cvetkovic	
Instructor	Dr. M. Cvetkovic	
Instructor	P. Manganiello	
Contact Hours / Week x/x/x/x	0/0/4/0 hc	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Course Contents	The course consists out of two parts: electricity supply and sustainable energy. In the first part students will be introduced to three-phase power systems and the most important components that constitute the power system. Students will also learn about the important challenges for integration of renewable energy in the power system. By attending this course, students will obtain basic knowledge of power systems which is needed to continue with further education in electrical energy systems (for example within Electrical Sustainable Energy Minor or within MSc in Electrical Power Engineering). In the second part students will learn how to estimate and calculate the potential of (renewable) energy sources in reference to humanity's energy consumption using the physical principles of energy generation and consumption. After completion of the course the student is able to design a sustainable energy infrastructure based on realistic calculations and conditions. In addition the student establish an understanding of the basic underlying electric engineering principles in - e-mobility, - solar energy, - wind energy and - high voltage/high power systems.	
Study Goals	- To learn how a power system works - to obtain basic knowledge how to model a power system with the associated components - to recognize the challenges of renewable energy integration and the engineering methods to solve them - to learn how a power system behaves during small and large disturbances - to obtain knowledge and techniques for solving power system problems (load flow, computation of short circuit fault currents) - To learn how to make simple estimations/calculations of the potential of (renewable) energy sources in reference to human kinds energy consumption using simple physical tools/concepts - to understand and apply the physics of energy generation and consumption - to calculate the energy generation and consumption - to design of a sustainable energy infrastructure based on realistic calculations - to understand the wide variety of electrical engineering principles in various examples of sustainable energy technologies - to establish an understanding of the basic underlying electric principles in - e-mobility, - solar energy, - wind energy and - high voltage/high power systems	
Education Method	Lecture/instruction material will be in advance provided on the Brightspace. Exercises will be solved during the lecture hours and will be provided on the Brightspace afterwards.	
Literature and Study Materials	Pieter Schavemaker, Lou van der Sluis, Electrical Power System Essentials David JC MacKay, Sustainable Energy - Without the hot air	
Assessment	The examination for this course will be conducted in two parts, specifically two written exams. The final score is the average of the two-part exam results. Hereby the calculation is done with one decimal place, and the final grade rounded to half. The re-examination is conducted in one part (all material), specifically one written exam. The result of the partial exams expires prior to resit and is not transferable to the next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE2L11	EPO-3: Design a Chip	5
Responsible Instructor	Dr.ir. A.J. van Genderen	
Contact Hours / Week x/x/x/x	0/8/0/0 prac	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	Dutch English	
Expected prior knowledge	Digital systems (EE1D11 and EE1D21), linear circuits (EE1C11 and EE1C21), Amplifier circuits and instrumentation (EE1C31), 1st year projects (EE1L11 and (EE1L21). Further, knowledge from the course integrated circuits (EE2C11), which is given in the first quarter, is heavily used.	
Course Contents	During this project, the student will learn how to make a design with a group. The project is a continuation of the projects during the first year of the bachelor program Electrical Engineering, but focusses more on structural hierarchical design and on working together within a group. The students are divided into groups of about 7-10 persons and each group together designs a chip.	
Study Goals	After finishing this course, the student is able to - make a design using global product specifications, including boundary conditions, - design in a systematic hierarchical way, - use analysis together with synthesis during the design process, - analyse his design at different levels of abstraction, - use different types of models during the design, - take into account testability, - perform the above design process within a group, - make a design as a part of a group design, - use modern computer tools during the design, - make documentation about the design.	
Education Method	project education	
Literature and Study Materials	Manual Project Design a Chip and other information on Brightspace	
Prerequisites	EPO-1 (EE1L11), Digital Systems A and B (EE1D11 and EE1D21) and lab of Integrated Circuits (EE2C11).	
Assessment	A grade is given for a module design report that is written halfway during the project, by 2 or 3 students, about a part of the design. A group grade is given for the total group design. This grade is based on the design itself and the report about it, the presentation and the defense. The group grade is adjusted by a peer review that is held. When G is the group grade, P is the possible adjustment made by a peer review ($-1 \leq P \leq 1$), and M is the grade for the module report, then the final individual grade is computed as $0.8 \times (G + P) + 0.2 \times M$.	
Judgement	A group grade is determined based on the evaluation of: - Originality and complexity of the design - Achieved results with respect to simulation and prototyping - Submitted design files - Test plan - Final report - Presentation and defense - Collaboration within the group The individual grade is then determined by increasing or decreasing the group grade by a maximum of 1 point, based on a peer-review within the group.	

EE2L21	EPO-4: "KIT" autonomous driving challenge	5
Responsible Instructor	Prof.dr.ir. A.J. van der Veen	
Instructor	Prof.dr.ir. A.J. van der Veen	
Instructor	B. Abdikivanani	
Contact Hours / Week x/x/x/x	0/0/0/8 prac	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	Dutch English	
Expected prior knowledge	The practicum applies knowledge of EE2S11 Signals and Systems, EE2S21 Systems and Control, the practicum of EE2T11 Telecommunication A, and EE2S31 Signal Processing (running in the same quarter). The "biosensing" variant assumes knowledge of EE2ML1 Introduction to Machine Learning.	
Parts	EPO4 offers two variants: "KIT: Autonomous driving" and "Biosensing".	
Course Contents	Variant "KIT": In this project you team with your fellow students to remotely operate an electric toy car called "KIT". The car is equipped with ultrasonic parking sensors, a Bluetooth connection, an audio beacon and a micro-controller. You are in charge of this "enhanced" car and will design and implement algorithms so that the car can recognize and execute your voice commands; it will be able to localize itself, sense obstacles and find its way autonomously; as well as to communicate with a base station at which calculations are performed. Reliable speech recognition, high tracking accuracy and speedy and tight control are your objectives. In a series of competitions you and your team will show your achievements in challenges of increasing difficulty. Variant "Biosensing": here you develop an Arduino-based stress detector, which involves both the hardware setup and data collection, and biosignal processing (machine learning) for stress detection. We use ECG sensors for recording heart rate, GSR sensors for recording skin conductance, and an Arduino Uno development board as the central controller for the system to record the data and store it on a computer. The signal processing part includes preprocessing of the signals to remove noise and artifacts, and machine learning tools for feature extraction to develop a classifier for stress and rest detection.	
Study Goals	The practical involves the design of a complex system. The system has both hardware and software components that interact with each other. The system consists of modules, each specified, designed and tested separately, and subsequently integrated into the overall system. The practical integrates knowledge acquired in the other BSc courses. By the end of the course, you should be able to: LO1. Analyze a real-world application which involves both hardware and software by breaking it down into smaller modules; LO2. Apply signal processing and machine learning (and control) tools to design and implement the solution using matlab or python programming language; LO3. Systematically validate the solution (per module and end-to-end); LO4. Plan, manage and conduct effective teamwork for the successful realization of a technical complex assignment. LO5. Write and present a technical report that complies with current standards of the electrical engineering scientific community.	
Education Method	Project oriented education. EPO-4 has 14 scheduled lab mornings. Interleaved with these there are scheduled preparation/homework mornings. In week 9 there is a common morning for the final challenge, and in week 10 you will be called for a final presentation followed by a discussion. The total labtime is 60 hours, preparation/homework time is budgeted at 80 hours, for a total study load of 5 ECTS. Visit the Brightspace page of the current course for the detailed working schedule.	
Literature and Study Materials	Brightspace is used as educational platform. The practicum manual, detailed course description and materials, and communication channels are available through the Brightspace website of the course.	
Practical Guide	A practicum manual and additional data sheets are available through Brightspace.	
Prerequisites	Admission is provided to students having fulfilled: 1. Practicum of EE2T11 Telecommunications A 2. EE1L21 EPO-2: Smart Robot Challenge The "Biosensing" variant of EPO4 requires in addition that you passed EE2ML1 Introduction to Machine Learning. If you do not (completely) fulfill the requirements, but you would like to be admitted to the project, you can indicate this together with your motivation in a form provided via Brightspace. Assessment to your request will be given after the submission deadline. Visit Brightspace for details.	
Assessment	Summative assessment is in the form of: (1) competitions or design review presentations; (2) midterm report; (3) final report; (4) oral presentation and defence.	
Enrolment / Application	Enrollment is required. Restrictions for enrollment apply regarding prior courses that should have been completed before taking this course. Enrollment uses an online form that is published on Brightspace approximately 2 weeks before the start of the course. Although you can specify your preference for either option (KIT or Biosensing), capacity for each option is limited so it is possible that you will be assigned to an option that you did not select.	
Remarks	Please check the rules and regulations that apply. These can be found in the manual and on Brightspace. In summary: 1. Presence during labdays is compulsory 2. Upon non permitted absence for more than two times, you will be removed from the practicum 3. Preparation for labdays is necessary 4. Deadlines are strict. If you hand in your deliverables after the deadline, this will affect your grade	
Judgement	The final group grade depends on the performance in the challenges, the judgement of the reports, and the final presentation/defense in front of a committee. Individual grades can be adjusted via a peer-review system and staff observations, which can lead to a maximum 2 point variation (-1 to 1) with respect to the average group grade. Remarks: Submitting your reports late will be negatively reflected on your grade at the lecturer's discretion. It is required that your mid-term report is graded with at least 5. If that is not the case, you will get a chance to improve the mid-term report and resubmit. Resit. If the final grade is insufficient, you will have the chance to improve your grade by improving and resubmitting any (or both) of your midterm and final reports.	

EE2M11	Complex Analysis	5
Responsible Instructor	Dr. J.A.M. de Groot	
Contact Hours / Week x/x/x/x	6/0/0/0 hc	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	First year Analysis, particularly having previous experience with topics such as complex numbers, limits of functions, LHospitals theorem, (partial) differentiation, diverse standard functions, integration, line integrals, Greens theorem, power series	
Course Contents	Complex numbers, properties of $\mathbb{C}$, complex functions, analytic functions, the Cauchy-Riemann equations, integrals of complex functions over curves in $\mathbb{C}$, Cauchy's theorem, Cauchy's integration formulas, Taylor- and Laurent-series, classification of singularities, Residue Theorem, contour integrals, Jordans lemma, Fourier- and Laplace integrals	
Study Goals	The student must become acquainted with a number of topics from Complex Analysis so that he/she will be able to apply them in the classes Signals and Systems and Electromagnetism.	
Education Method	Four hours online lecture and 2 hours online instructions (discussing the material and solving problems) per week	
Literature and Study Materials	Since 2011-2012, all material will be taken from Complex Variables and Application, written by J.W. Brown and R.V. Churchill. Aside from this, the instructor will provide detailed lecture slides that will be uploaded episodically onto Brightspace throughout the course.	
Assessment	A two-part examination. Also see Judgement	
Remarks	During the course, the instructor will upload lecture slides, explanations, assignments with sometimes solutions, announcements, lecture hall changes, old exams with solutions, and more onto Brightspace. Make sure you check the course EE2M11 on Brightspace regularly!	
Judgement	The exam is split into two parts, one in week 5 and one in week 10, both last two hours. Your final grade is then equal to the average of those two parts. The grades of the parts are not rounded; the final grade is rounded to the nearest half. The exam resit has only one part, taking material from the entire course (you can not resit only one part of the original exam). Your score on the resit overrides the scores of the two original parts, and the scores of the original two parts is not transferrable to the following academic year. The exams contain open questions and possibly some short answer questions.	
Co-Instructor	Drs. W.H.H. de Muinck Keizer	

EE2M21	Linear Algebra and Differential Equations	5
Responsible Instructor	Drs. A.T. Hensbergen	
Contact Hours / Week x/x/x/x	0/6/0/0 hc	
Education Period	2	
Start Education	2	
Exam Period	2 5	
Course Language	English	
Expected prior knowledge	Highschool math; Linear Algebra 1 (1st year course)	
Course Contents	Linear algebra and differential equations	
Study Goals	Linear Algebra: 1. Determinants. 2. Eigenvalues en eigenvectors. 3. Decomposition of matrices, diagonalization and QR-decomposition. 4. Orthogonality in $\mathbb{R}^n$: orthonormal bases, orthogonal matrices, Gram-Schmidt process. Differential Equations: 1. Linear differential equations (first and second order, special second order and special methods). 2. Systems of linear differential equations. 3. Non linear Systems of non-linear differential equations; stability. 4. Fourier series. 5. Partial differential equations; the method of separation of variables.	
Education Method	Lectures and tutorials Apart from that there are exercises available (in GraspLe).	
Assessment	1. In week 5 and in week 10 there are two exams of 2 hour. 2. The resit is an 3 hour exam. 3. In case corona is still around, so on campus examinations are prohibited, the exam(s) will take place via Brightspace and/or GraspLe. For up-to-date information Brightspace is the place to be.	
Exam Hours	Two partial exams: 2 hours; resit: 3 hour	
Permitted Materials during Tests	Pen and pencil.	
Judgement	1. The final result is the average on the results of the exams in week 5 and week 10. 2. The final result of the resit is purely based on the resit, (partial) results from the past are not taken into account.	
Co-Instructor	Dr. F.J. van de Bult	

EE2S11	Signals and Systems	5
Responsible Instructor	Prof.dr.ir. A.J. van der Veen	
Instructor	Dr.ir. R.F. Remis	
Contact Hours / Week x/x/x/x	0/4/0/0 hc	
Education Period	2	
Start Education	2	
Exam Period	2 5	
Course Language	English	
Required for	EE2S31 Signal processing, EE2S21 Systems and control, EE2T11 Telecommunication A, EE2L21 KIT Autonomous driving challenge	
Expected prior knowledge	EE2M11 Complex Functions	
Course Contents	Starting from complex function theory, the course presents the theory of signal transformations and linear time-invariant systems, preparing for courses on signal processing, systems theory, and telecommunication. Part 1: continuous-time signals - Linear time-invariant systems (LTI), eigenfunctions and eigenvalues of an LTI system, Dirac delta function, Heaviside unit step function. Convolution. - Laplace transform, region of convergence. Properties. Convolution and product, inverse transform. - Periodic signals. Fourier series, line spectrum. - Fourier transform (continuous time). - Transformations of a pulse, sinc function, sign function, step function. Duality of the Fourier transform. Shift and modulation, Parseval's theorem. Part 2: discrete-time signals - Ideal sampling, impulse train, periodic spectrum. Sampling theorem (Nyquist, Shannon). Ideal reconstruction - Discrete signals and systems, convolution. - Z-transform - Realizations (direct form for FIR, IIR) - Fourier transform (discrete time) - Filter design (analog and digital), frequency transformations.	
Study Goals	Understanding and applying integral transformations that play a role in LTI signals and systems.	
Education Method	Lectures, combined with a few practice sessions. (A practical on the course contents is included as part of EE2T11 Telecommunicatie A.)	
Books	"Signals and Systems using MATLAB, Third Edition" by Luis Chaparro and Aydin Akan, Academic Press; 3rd edition (2018). ISBN: 978-0128142042.	
Assessment	Written exam, with a first test in week 5 (part 1) and a second test in week 10 (part 2). Resit in summer (complete)	
Permitted Materials during Tests	Closed book; one A4 page (2 sides) with notes permitted	
Judgement	The exam is taken in two parts. The final mark is the average of the marks of each part. The partial marks are determined with an accuracy of 1 decimal, the final mark is rounded to half a digit. The resit is a single test on the complete course. The results of the partial tests are not used for the resit, and are not valid for the next year.	

EE2S21	Systems and Control	5
Responsible Instructor	Dr.ing. S. Grammatico	
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Expected prior knowledge	* Roots of quadratic functions and higher-degree polynomials * Linear differential equations * Laplace transforms * Partial fraction expansions * Complex numbers * Dynamics of electrical and mechanical systems	
Course Contents	We study properties of dynamic systems and how these properties can be influenced via control. To this aim we first develop mathematical models for the dynamic behavior of systems based on first principles. Next, we analyze several important system properties in the time and the frequency domain. We introduce the concept of feedback control and discuss its effects on the system properties. Finally, we present several feedback control approaches to influence the system properties.	
Study Goals	At the end of the course, the students should be able to 1. to describe simple physical systems (e.g. electrical, mechanical, electro-mechanical systems) via mathematical dynamic models; 2. to analyze the dynamic behavior of closed-loop systems described by state-space models or transfer functions in terms of step and impulse responses, zeros and poles; 3. interpret Bode diagrams of 1st and 2nd order systems, and sketch Bode diagrams of systems given via a transfer function; 4. design controllers for linear dynamical systems; 5. apply the above approaches and methods via dedicated software (e.g. Matlab).	
Education Method	Lectures, instructions, lab, self-study.	
Literature and Study Materials	Feedback Control of Dynamic Systems, by G.F. Franklin, J.D. Powell, and A. Emami-Naeini, Seventh Edition, Prentice Hall, 2014.	
Assessment	Written exam.	
Tags	Lineair Algebra Mathematics Signals and Systems	
Judgement	Written exam.	
	Final grade = grade for the written exam. This also holds for the resit.	
Co-Instructor	Dr.ing. S. Grammatico	

EE2S31	Signal Processing	5
Responsible Instructor	Prof.dr.ir. A.J. van der Veen	
Instructor	Prof.dr.ir. A.J. van der Veen	
Instructor	B. Hunyadi	
Contact Hours / Week x/x/x/x	0/0/0/6 hc	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Continuation of EE2S11 Signals and system, EE1M31 Probability and statistics; Uses EE1M21 integration and differentiation.	
Course Contents	Signal processing plays an important role in many applications, such as consumer electronics (audio and video applications), radar and communication, and biomedical applications. It is one of the major areas in Electrical Engineering. This course consists of two parts: digital signal processing and stochastic processes. The two parts have separate books and will be taught in parallel. The part on digital signal processing covers one-dimensional signals and discuss non-ideal sampling, DFT and FFT, spectral analysis, multirate filters, and (quantization) noise propagation in filters, and sigma-delta modulation (ADC). The part on stochastic processes introduces stochastic models and random processes to describe systems and signals that are not deterministic in nature. In this course the following topics are discussed: Signal processing: 1. Repetition ((discrete-time) signal processing, z-transform, poles and zeros, filter functions) 2. Non-ideal sampling and reconstruction 3. Discrete Fourier Transform (frequency domain sampling, properties, convolution using DFT (FFT)) 4. Spectral analysis 5. The FFT 6. Multirate filters 7. Quantization and rounding errors 8. Sigma-Delta modulation (ADC) Stochastic processes: 1. Repetition random variables. Pairs of random variables. Correlations. 2. Random vectors. Conditional probability models 3. Sums of random variables; the moment generating function. Performance bounds: Markov, Chebyshev, Chernoff inequalities. Bias, consistency. 4. Estimation of random variables (MMSE, MAP, ML) 5. Stochastic processes, autocorrelation function, stationarity, wide sense stationarity, estimation of the correlation function 6. Signal processing of stochastic signals, power spectral density functions, white noise. 7. AR processes, linear prediction.	
Study Goals	After passing this course the student is able: 1. To show the relationships between analog and discrete-time spectra of signals, and the effect of upsampling and downsampling on these spectra; 2. To reason on (non-ideal) sampling and reconstruction, 3. To apply the theory of the DFT, and relate this to the DTFT, 4. To understand the background of the FFT, 5. To analyse the effects of coefficient quantization and roundoff errors in simple linear filters. 6. To work with and derive PDFs, PMFs, CDFs, expected values and auto-correlation functions on the basis of a problem description with a stochastic nature, 7. To calculate for a given probability model with multiple random variables the different (marginal, joint and conditional) parameters characterizing the model (eg. PDF, PMD, CDF, expected value, auto / cross-correlation function) 8. To determine and apply for random variables and random processes the following properties: independence, orthogonal, uncorrelated, IID. 8. To calculate the expected value, variance, auto/cross-correlation function and (cross) PSD for linear combinations of random variables and linearly filtered processes. To calculate and use the Moment Generating Function. 9. To apply the central limit theorem and the law of large numbers. 10. To estimate parameters describing statistical models based on realizations (based on MMSE, MAP, ML). 11. To transform certain electrical engineering problems into a stochastic model.	
Education Method	Lectures	
Books	Stochastic processes: R.D. Yates and D.J. Goodman, "Probability and Stochastic Processes", 3rd edition (International Student Version), and the (online) Supplement for this book; Signal processing: J.G. Proakis and D.G. Manolakis, "Digital signal processing, principle, algorithms and applications", 4th edition (Pearson international edition)	
Assessment	The examination of the course is conducted in two parts (in the middle and at the end of the quarter). Both subtests consist of a 50/50 distribution of questions on signal processing and stochastic processes, and the grade of each subtest is rounded to one decimal place. The final grade is the average of the two exam results, rounded to a half. The re-examination covers all lecture material. The result of the partial exams expires for re-examination and is not transferable to the next year.	
Co-Instructor	During the exam a hand-written formula sheet of 2 sides of an A4 page may be used. B. Hunyadi	

EE2T11	Telecommunications A	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Contact Hours / Week x/x/x/x	0/0/2/0 hc; 0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	Telecommunications B, EE2T21, part Telecommunication Techniques	
Expected prior knowledge	Analysis (in particular integration, differentiation and complex calculation). Signals and Systems: LTI-systems, convolution, impulse response, Fourier-transformation and filtering. Probability and Statistics: probability model, conditional probability, independence, probability density function (PDF), expectation and variance, auto-correlation and power spectral density (PSD) function.	
Course Contents	Transport of information on the basis of telecommunication forms the backbone of our society. According to a binding international treaty among more than 180 countries, "telecommunication" covers every transfer, transmission or reception of signals, tokens, sounds, images or other data of any kind (including ideas), by means of electromagnetic systems, e.g. via radio-electric or optical means. This concise legal definition covers a.o. speech and data communication, broadcast, electronic positioning and navigation, telemetry and remote sensing (tele-observation). In the course Telecommunications A all of these applications are reduced to simple "archetypal" communication systems and processes, represented by a block diagram, composed of more or less idealized subsystems as transmission path, filters, amplifiers, (de)multiplexers, (de)modulators, etc. These can be described with the prerequisite knowledge from the BScEE curriculum, especially Signals and Systems (EE2S11) and Probability and Statistics (EE1M31), but they will also need to meet external design requirements, for example on the quality and capacity of the system. This course, therefore, fulfills a bridging function between the abstract knowledge of the first years of study and the more task-based reality of the advanced engineering study. In Telecommunications A, the emphasis is on basic baseband systems and processes, and focuses on the physical properties of simple transmission media and noise, as well as the associated system trade-offs that can be made. In addition to the lectures, the course Telecommunications A (EE2T11) includes Matlab based lab course assignments (of 2 EC) in which skills related to Signals & Systems are further developed (convolution, Fourier transformation, filter design, deconvolution, sampling). The results of the deconvolution assignment are explicitly needed for the part localization using a microphone-array of EPO-4 (EE2L21).	
Study Goals	The student has gained insight in and understanding of the following concepts: - decibel notation, - mathematical description of signals and definitions of bandwidths, - propagation loss for guided (cable) and unguided (wireless) media, - noise and system noise calculations (link budget), - signal sampling, - analogue pulse modulation (PAM, PCM, Delta modulation), - digital signal waveforms in baseband, their power spectral density function and spectral efficiency, - line codes, and, based on this, is able to solve system related problems supported by calculations. In the course labs, the student has further developed his/her skills related to the concepts of Signals & Systems, can apply these concepts in Matlab scripts and can report on his/her findings in a concise and accurate way.	
Education Method	Lectures, instruction and working lectures, Möbius homework assignments and lab tasks. The course labs and the telecommunications lectures are given in two blocks in quarter 3: - block 1: course labs, during lecture weeks 3.1 - 3.5, - block 2: telecommunication lectures, during lecture weeks 3.5 - 3.10.	
Literature and Study Materials	Couch, L.W., Digital and Analog Communication Systems, 8th edition, ISBN 978-0-13-291538-0, Prentice Hall, 2013. Lab course manual.	
Assessment	To support the study of the material of this course, weekly Möbius homework assignments (seven in total) are made available via Brightspace. These assignments are graded and provide insight in how well the course material is mastered. The average grade (HW) for the homework assignments counts as part of the final grade. The lecture part of this course (2nd block) is completed with a 3 hour written exam covering the full course content at the end of the 3rd quarter or in the resit period.	
Permitted Materials during Tests	The exam Telecommunications A is a closed-book exam. During the exam, you may use the following materials: - the overview Tables and Figures to be used at the exam Telecommunications A (EE2T11) as made available on Brightspace, - an original own handwritten formula sheet: 1 page A4, written on both sides, (not type-written, no copy and it should not contain worked out exercises or examples), - a non-programmable electronic calculator.	
Enrolment / Application	To take part in the lab courses belonging to this course, you need to enroll yourself at least one week before the start of the lectures via the Brightspace page of this course.	
Special Information	This study guide information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Judgement	The final grade of Telecommunications A is determined based on the grade obtained for the exam or resit (TT) and the average grade for the Möbius homework assignments (HW). The partial grades are calculated with one decimal accuracy and the final grade is rounded to the nearest half. The grade for the course Telecommunications A is determined as follows: $\text{Grade} = \text{TT} + \text{HW} \cdot (10 - \text{TT}) / 100.$ The average grade obtained for the Möbius homework assignments (HW) is only valid in the year of study in which this result was obtained, so up to and including the resit exam. For a valid final grade for the course Telecommunications A, both the course labs and the exam need to be completed with a pass result. The course lab result will expire as a partial result at the end of the year of study in which it was obtained. This means that in case of an insufficient exam result (or when you did not take part in the exam or resit) in the same year of study, the course labs have to be done again in a next year of study.	
Co-Instructor	Prof.dr.ir. A.J. van der Veen	

EE2T21	Telecommunications B	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Contact Hours / Week x/x/x/x	0/0/0/4 hc	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Telecommunications A (EE2T11); Analysis (in particular integration, differentiation and complex calculation); Signals and Systems: LTI-systems, convolution, impulse response, Fourier transformation filtering and Hilbert transformation; Probability and Statistics: probability model, conditional probability, independence, probability density function (PDF), cumulative probability density function (CDF), expectation and variance, (auto-)correlation, power spectral density (PSD) function, Gaussian (Normal) distribution, central limit theorem.	
Course Contents	Transmission and distribution of information by means of telecommunication techniques and networks form the backbone of our society. As a follow-up on Telecommunications A (EE2T11), the following aspects will be treated in Telecommunications B: Part 1. Telecommunication Techniques: - modulation techniques to convert analogue and digital information bearing signals onto signals suitable for transport over a guided (cable) or unguided (wireless) medium, - quality of detection of these signals in the presence of disturbance due to thermal noise, - related system trade-offs that can be made. Part 2. Data Communications Networking: this part of the course provides an introduction to telecommunication networks, which includes local area networks, the Internet, and telephone networks.	
Study Goals	The student has gained insight in and understanding of the following concepts and, based on this, is able to solve system related problems supported by calculations. In the part Telecommunication Techniques, the student will learn: - bandpass signals and systems: complex equivalent baseband description, modulation, demodulation - generic transmitter and receiver concepts, - linear analogue modulation techniques (AM, DSB, SSB), - non-linear analogue modulation techniques (PM, FM), - digital modulation techniques (OOK, FSK, BPSK, QPSK, MSK, DS-SS and OFDM), - signal-to-noise ratio before and after detection of analogue signals, - matched filtering and bit error probability after detection of digital signals. In the part Data Communications Networking, the student will learn to: 1. Explain the functionalities with which modern end-to-end communication in data communication networks is achieved. 2. Explain the architecture and design concepts that underlie the internet and its protocols. 3. Perform elementary calculations and performance analyses on small networks. 4. Implement basic mathematical coding and routing algorithms that have inspired error correction methods and routing protocols in networks.	
Education Method	Lectures, working and instruction lectures and homework assignments. The parts Telecommunication Techniques and Data Communication Networking are scheduled as block lectures: - Telecommunication Techniques: during lecture weeks 4.1 - 4.5, - Data Communication Networking: during lecture weeks 4.6 - 4.10. To support the study of the material of Telecommunication Techniques, weekly Möbius homework assignments (seven in total) are made available via Brightspace. These assignments are graded and provide insight in how well you master the course material. The average grade (HW) for these homework exercises (0 - 10) counts as part of the final grade. For Data Communications Networking, (a) bonus assignment(s) will be provided and graded. The points scored on these assignment(s) counts towards the final grade.	
Literature and Study Materials	Couch, L.W., Digital and Analog Communication Systems, 8th edition, ISBN 978-0-13-291538-0, Prentice Hall, 2013. P. Van Mieghem, Data Communications Networking, Chapters 1-7, 2011, ISBN: 978-94-91075-01-8.	
Assessment	The part Telecommunication Techniques is completed with a written partial exam (result TT1) in week 4.5 or during the resit period (result TT2). The part Data Communications Networking is completed with a written partial exam in the exam period of Q4 (result TN1) or in the resit period (result TN2).	
Permitted Materials during Tests	The partial exam Telecommunication Techniques is closed-book. During this exam, you may use the following materials: - a non-programmable electronic calculator, - the overview Tables and Figures to be used at the exam Telecommunications B (EE2T21), as made available on Brightspace, - an original self-handwritten equation form: 1 page A4, written on both sides, (not type-written, no copy and it should not contain worked out exercises or examples). The partial exam Data Communication Networking is open-book.	
Special Information	This study guide information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Judgement	The final grade of the course Telecommunications B is determined as follows based on the partial grades of the initial exams {TT1 en TN1}, the resit exams {TT2 and TN2} and the grades obtained with the homework exercises {HW, B}: - Grade Telecommunication Techniques: $CT = TT + HW (10 - TT) / 100$, - Grade Data Communications Networking: $CN = TN + B (10 - TN) / 100$, where $TT = \max\{TT1, TT2\}$ and $TN = \max\{TN1, TN2\}$. For a valid final grade, the partial grades CT and CN should be both sufficient (at least 5.8); then the final grade for Telecommunications B is found as: - Final grade = $0.6 CT + 0.4 CN$. The calculation of the partial grades is done with an accuracy of one decimal and the final grade is rounded to the nearest half value. When a part is completed after the first exam opportunity with a sufficient grade (CT or CN) no resit of this part is needed. When the course is not completed after the resit (i.e. when either or both of the partial grades CT and CN are insufficient), then all obtained partial grades as well as the results obtained for the homework assignments will expire. Thus, none of the results can be transferred to a next study year.	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Bachelor Electrical Engineering

B-EE jaar 3 2022	
Director of Studies	Dr.ir. N.P. van der Meijs
Program Coordinator	H. Shekarian Tabarestani
Program Coordinator	L.M. Ramirez Elizondo
Prerequisites	Let op! De Ingangseisen zijn in de OER vastgelegd. Mocht er verschil zijn tussen wat er hier staat en wat in de OER staat, dan is datgene wat in de OER staat geldig. Om deel te mogen nemen aan het studieonderdeel EE3L11 Bachelorafstudeerproject Electrical Engineering moet je alle vakken van het eerste en tweede jaar van het je bachelorprogramma behaald hebben.
Introduction 1	Inschrijven Voor EE3L11 Bachelorafstudeerproject Electrical Engineering dien je je in te schrijven. Meer informatie vind krijg je via een announcement via Brightspace en tijdens de informatiebijeenkomst in het 3e kwartaal. Voor de (deel-)tentamens van alle vakken moet je je uiterlijk 1 week vóór de dag van het tentamen via OSIRIS inschrijven. Wil je bij nader inzien toch niet meedoen aan een studieonderdeel of tentamen waar je je voor hebt ingeschreven? Schrijf je dan ook weer uit!
Introduction 2	Zie voor uitleg https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens In het derde jaar volg je een minor. Kijk voor meer informatie op http://minors.tudelft.nl. Als je na je bachelor nog een master wilt volgen, dan is het nu tijd om je te oriënteren. Er wordt twee keer per jaar een masterevent georganiseerd door de TU Delft. Tevens kun je kijken op doorstroommatrix.nl voor welke opleidingen je toelaatbaar bent.

EE3C11	Electronics	5
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Contact Hours / Week x/x/x/x	0/0/6/0 hc	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Course Contents	This course has two parts: Electronics and Device Physics. The Electronics part deals with structured design of analog electronic circuits. The design method presented in this course is based on network theory, control theory, signal processing and physics. The design of analog electronic circuits is considered by many people to be complex. This is mainly because designers have to deal with many relevant performance aspects that can be achieved in many different ways. In other words, there are many degrees of freedom for obtaining the desired performance of an electronic circuit. Theoretical concepts, circuit topologies, electronic devices, their operating conditions and the physical lay-out of a system, together constitute an enormous design space in which it is easy to get lost. For this reason, analog electronics is often regarded as an art rather than a solid discipline. Instead of taking numerous existing solutions to known problems as a starting point, it will be shown in this course that it is far more effective to start a new design with clear distinction between theoretical concepts and study how they can be implemented in such a way that the possibilities of the implementation technology are maximally exploited. Such an approach not only results in clear and reliable designs with predictable performance, but it also provides a basis for sharing and developing knowledge. During the engineering phase, designers need to find real world components, determine their operating conditions and fix these conditions by design. To this end they need to formulate and solve so-called design equations for the circuit. For example, in order to determine a budget for the gain-bandwidth product of an operational amplifier, we must find relations between this gain-bandwidth product and the performance requirements of the circuit that comprises this operational amplifier. Setting up design equations for analog circuits, even for circuits based on operational amplifiers, is a difficult and cumbersome task. It requires knowledge about technology, modeling techniques and interactions between the behavior of the circuit and the properties of the operational amplifier. It goes hand in hand with symbolic analysis. For this, during this course we will use SLiCAP (Symbolic Linear Circuit Analysis Program). SLiCAP is written in MATLAB®. Although the method is applicable to many different electronic systems, the focus in this course is on the design and implementation of high-performance operational amplifier circuits. (Transistor level details will be discussed in the master course EE410a9.) The physics part of the course concentrates on the physical description of the most relevant transistors (bipolar and MOS based). Device models play an important role in the translation of an ideal electrical network into an electronic circuit. These models contain parts that match the required ideal network theoretical behavior and parts that model the physical limitations. Electronic design is basically nothing more than finding the optimal electrical network that meets the system requirements and then translating it into a network that is physically realizable by gradually refining the models used and keeping the influence of non-idealities that are introduced to a minimum in the process. It is important for professional designers to understand the physical background of the device models to be able to estimate the accuracy and the stability of the parameters. In the end this gives the expected performance of the circuit in the real physical world. It also gives clues for further technology improvement. It is important to know which parameters are most dominant in restricting the circuit performance. It is needed to select the technology that is best suited to meet all requirements or prove that the requirements are beyond (present) physical limits.	
Study Goals	After the course the student is able to: - translate an application description of an amplifier into a specification comprising, a functional specification, specification of performance measures, operational conditions and cost factors. - design a proper amplifier concept based on source and load specification. - identify performance aspects such as noise, signal handling capability and bandwidth of basic of operational amplifiers. - select operational amplifiers have sufficient specifications for the amplifier being designed and select their operating conditions to influence these performance aspects. - to design (high-performance) operational amplifier based amplifiers using error reduction techniques such as compensation, balancing and feedback. - setup design considerations for noise and take adequate design decisions. - setup design considerations for signal handling capability and power efficiency and take adequate design decisions. - setup design considerations for frequency response and take adequate design decisions. - setup design considerations for weak distortion and take adequate design decisions. - setup design considerations for biasing and design and implement proper biasing circuitry. - evaluate the performance versus requirements of the final design and propose options for improvement. - select proper models and design verification tools for each design step. - write a design report in which all design steps are motivated and the performance is evaluated and explained. - name and use the basic concepts from semiconductor physics when explaining the functioning of a PN junction. - explain the properties of bipolar and cmos transistors with the corresponding electrical models from the physical phenomena in the transistors	
Education Method	Lectures, assignments, instructions	
Computer Use	It is necessary to have a computer available with SLiCAP installed and the free circuit simulator LT-Spice for the electronics part of this course.	

Assessment	SLiCAP is an acronym for: Symbolic Linear Circuit Analysis Program. During the first lecture, the installation and use of SLiCAP will be explained.
	See: https://www.analog-electronics.eu/slicap/slicap.html
	An exam composed of an electronics part and a device physics part. The average of the grades of the electronics part and the device physics part is the final grade of the exam.
Co-Instructor	Depending on the availability of proper facilities and logistics, it may be made possible to earn the grade for the electronics part by designing, building, and testing a circuit using the knowledge and skills provided in this course. In that case, the assessment for the electronics part is a design review. More details will be provided at the start of the course.
Co-Instructor	Dr. R.A.C.M.M. van Swaij
	A.J.M. Montagne

EE3D11	Computer Architecture and Organisation	5
Responsible Instructor	Prof.dr.ir. S. Hamdioui	
Instructor	Ing. A.M.J. Slats	
Instructor	M. Taouil	
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Expected prior knowledge	Digital systems or equivalent course on logic design Basic concepts of programming in C	
Course Contents	This course provides an overview of the architecture and organization of a computer hardware system and the important principles of computer organization. In addition, the course addresses also object-oriented programming techniques. The course demonstrates the interrelation between hardware and software, and illustrates how the computers operates and how they can be programmed, with the emphasis on processor design and implementation. Topics discussed are Computer system overview, Measuring and comparing performance, ISA: instruction set architecture (MIPS, x86, 8051, JVM), Computer arithmetic, processor implementation, fast processor implementation, Memory hierarchy and caching, Interfacing, Modern architectures and organizations (superscalar, VLIW/IA64/Itanium), etc.	
Study Goals	The main aims of the course are: 1. Understand the basics of computer abstraction and performance: what are the different computers? How to evaluate performance? What is the impact of technology scaling? Etc. 2. Understand and use the computer language: What is an ISA and how to write and extend it? How does it support procedures in hardware? Impact of language selection of hardware design? Etc. 3. Understand and perform computer arithmetic: how to perform addition, subtraction, multiplication and division for integers and floating point? What is the impact of the algorithm on hardware implementation? Etc. 4. Understand and use the principle and techniques used to design a processor: how does hardware execute instructions? What are the performance characteristics and limitations? How to accelerate the performance? Etc. 5. Understand the memory hierarchy and its impact on performance: What is the memory hierarchy? What are the characteristics and limitations of each hierarchy level? What are the different alternatives to improve the performance? Etc. 6. I/O systems and their interaction with the processor: what are the different I/O? How does the communication take place? How are the dependability, reliability and availability requirements for each I/O? etc. 7. Understand and describes the trends and the challenges in the field of computer architecture. 8. Use object-oriented programming techniques in C++ such as: classes and objects (definition and implementation) and inheritance (types, multiple inheritance, redefinition).	
Education Method	Lectures + lab	
Literature and Study Materials	Lecture Notes Computer Organization & Design MIPS Edition: The Hardware/Software Interface, by David A. Patterson & John L. Hennessy; 6th Edition; Morgan Kaufmann E-book "C++ Language Tutorial," Juan Soulie, www.cplusplus.com/doc/tutorial	
Assessment	The assessment consists of the following parts: 7 Lab assignments: Pass/Fail 5 homework: HW1, HW2, HW3, HW4 and HW5 (weekly). 1 written exam: WE	
Judgement	Final grade of the first exam If [(Lab passed) AND (HW>=6)] then: 'Finale grade' = max((0.20*HW + 0.80*WE), WE) Where HW is the weighted average grade of all homework; HW = (HW1 + HW2 + HW3 + 2*HW4 + 2*HW5)/ 7. WE is the grade of the written exam. Final grade of retake exam: If (Lab passed) then: 'Finale grade' = WE	
Co-Instructor	M. Taouil	

EE3L11	Bachelor graduation project Electrical Engineering	15
Responsible Instructor	Dr.ing. I.E. Lager	
Instructor	Ir. E.W. Bol	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Prof.dr. C.S. Richie	
Contact Hours / Week x/x/x/x	0/0/0/5 project (0/5/0/0 project)	
Education Period	1 2 4	
Start Education	1 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Access to Bachelor graduation project is only granted upon fully completing: the curriculum for the first two years for access to the main entry point; all credits in the Bachelor curriculum, except for those for EE3L11, for access to the alternative entry point. The reference moments are: the end of the first semester for the main entry point; the beginning of the first semester for the alternative entry point.	
Course Contents	The Bachelor graduation project is a complex educational activity offering the students the possibility to materialise the knowledge and practical skills accumulated during the complete Bachelor curriculum in a design project in the area of Electrical Engineering. The students are expected to go through the complete design process, from drawing up a set of requirements, through the concept analysis and design, concept selection, to the presentation of the final design, in a structured and iterative manner. For stimulating an entrepreneurial spirit, the graduation project also requires the students to write a business plan describing how the technology can be transformed into some commercial product. In view of the prime importance of the societal awareness for the future engineers, the students will be also involved in an Ethics and technique module that will discuss responsibility, ethical theories, ethical dilemmas in companies, taking decisions in an environment of uncertainty, ethics and law etc. The modules expected effect is to train students to think about their role as future engineers and be able to raise their voice and participate in moral debates in their future occupation. The project is to be carried out in groups of 6 students (groups consisting of less students may also be accepted in special cases). The groups task is approached concurrently by 3 or 2 sub-groups that address specific aspects or components of the given assignment. Each sub-group is required to draw a thesis (one per sub-group) that must then be defended in front of a jury consisting of specialists in the area (see below). The complete group must also jointly prepare one Business Plan that must be presented and advocated during a competition termed Grand Finale. The project also contains a mandatory, advisory mid-term evaluation denoted Green-light assessment and meant for rating the readiness of each sub-group to draw a defensible thesis at the regular completion deadline. The Bachelor graduation project is carried out, as a rule, in the fourth quarter of the third year (the main entry point). Access to this discipline is only granted upon fully completing the curriculum for the first two years. For those not reaching the entry requirements at the reference moment (see below), a second round will be scheduled during the second quarter of the next year (the alternative entry point). The entry requirement for this point is the complete Bachelor curriculum, except for EE3L11. The reference moments are: the end of the first semester for the main entry point; the beginning of the first semester for the alternative entry point. This discipline consists of 3 modules: Bachelor thesis, Business Plan and Ethics and technology. The credits are granted as follows: 10 ECTS for the Bachelor thesis itself, 3 ECTS for the Business Plan module and 2 ECTS for the Ethics and technology module.	
Study Goals	Apply, integrate and deepen the knowledge accumulated in the previous study years. Account for the societal facets of the Electrical Engineering. Cooperate in a group for resolving a complex, open task. Identify the pertaining relevant literature sources and assess their suitability. Devise and articulate a design process according to the generally accepted procedures. Develop a prototype. Report on a technical subject in a scientific manner that also highlights the development process. Present a technical subject in a clear manner to a professional audience. Experience the various aspects involved by starting up a company. Write a (relatively simple) business plan and present/advocate it convincingly. Critically and pro-actively analyse and address the societal facets of industrial activity, with an emphasis on those occurring in the realm of electrical engineering. Recognise moral dilemmas in the vocational practice. Be able to judge moral issues with the instruments of ethical reflection.	
Education Method	Design project in a team. For the "Business plan" and "Ethics and technology" there will be courses and instructions for which presence is mandatory.	
Assessment	The assessment is done separately for the 3 modules: (1) Bachelor thesis: The students must submit a Bachelor graduation thesis, one per sub-group. The thesis must be defended per sub-group in front of a graduation commission appointed by the faculty, the assessment of the commission being for each student, individually. (2) Business Plan: The complete group must draw a Business Plan. The plan must be defended per group in front of a commission, the assessment of the commission being for the complete group. The business plan is defended during a special event, the Grand Finale, that concludes EE3L11. (3) Ethics and technology: The students must prepare an essay that, together with the students' activity during workshops, yields the grade. The essay is prepared per groups. Students not obtaining a passing grade will be requested to do an oral resit that is organised by appointment.	

Judgement	The final grade is computed as $G = 10/15 \cdot B_t + 3/15 \cdot B_p + 2/15 \cdot E \& t$ where B_t is the partial grade for Bachelor thesis, B_p the one for Business Plan and $E \& t$ the one for Ethics and technology. The minimum for all partial grades is 6.0.
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EE3P11	Electromagnetics	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	O.A. Krasnov	
Contact Hours / Week x/x/x/x	0/0/6/0 hc; 0/0/6/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Expected prior knowledge	EE1M11/21/31 Linear algebra and calculus (differentiation, integration, complex numbers, Euler's formula, vector algebra, vector calculus, gradient, divergence, curl, integral theorems for gradients, divergences (Gauss's integral theorem), and curls (Stokes's integral theorem)), EE1P21 Electricity and Magnetism (electrostatics and magnetostatics), EE1C11/21 Linear Circuits (Kirchhoff's laws, resistance, capacitance, inductance), EE2S11 Signals and Systems (the Fourier transform), EE1C11/21 Complex function theory and differential equations (principle branch of the square root and simple differential equations)	
Course Contents	Review of vector algebra and calculus, electrostatics, magnetostatics, Lorentz's force law, Maxwell's equations in vacuum, the compatibility relations and causality, conservation of charge, time-harmonic electromagnetic fields, polarization state, Maxwell's equations in matter, induced currents and the constitutive relations, the electromagnetic boundary conditions, Poynting's theorem for transient and time-harmonic waves, uniform and nonuniform plane waves, TE and TM electromagnetic waves, reflection and transmission at planar interfaces, the Fresnel reflection and transmission coefficients, Brewster angle and total reflection, the electromagnetic field in a highly conducting material, eddy currents, skin effect, induced surface current and surface impedance, transverse electromagnetic waves (TEM waves), transmission lines and the basic transmission line equations (telegraph equations), characteristic impedance, the coaxial line and the parallel-plate waveguide, propagation along lossless and lossy transmission lines, introduction to radiation, generalization of Coulomb's law and the Biot-Savart law for time-dependent fields	
Study Goals	After a successful completion of this course, you will be familiar with electromagnetic fields and electromagnetic wave propagation in a variety of settings as encountered in Electrical Engineering and other fields. You will have a thorough understanding of Maxwell's equations and you will be able to explain in your own words how electromagnetic waves propagate through different materials. Moreover, you will know and understand the different kinds of reflection and transmission effects that can take place at the interface between two different media. You will also be able to predict what types of induced phenomena (such as heat production, charge accumulation, and surface current generation) can take place within a piece of matter or at its boundaries. In addition, you will have an understanding of how the electromagnetic field can be used to transport energy (information) from one point to another and you will recognize that so-called transmission lines can be used to guide this energy transport. Finally, you will know how the voltages and currents along these lines are related to the electromagnetic field quantities and you will have a firm knowledge and understanding of the propagation properties of these lines.	
Education Method	Lectures and tutorials	
Literature and Study Materials	Book: Fundamentals of Applied Electromagnetics, F.T. Ulaby, E. Michielssen, and U. Ravaioli, 7th edition, Pearson. Slides: Available on Brightspace.	
Assessment	Written exam and lab sessions	
Permitted Materials during Tests	A one-page summary (handwritten, two sides of one A4) and a (graphic) calculator.	
Judgement	The course EE3P11 Electromagnetism consists of lectures and two obligatory lab sessions. Your final grade is computed as follows: $T =$ your grade for the exam or resit exam, $L =$ your grade for the lab sessions. If $T \geq 5,0$ and $L \geq 6,0$ then your final grade = $0.85 \cdot T + 0.15 \cdot L$.	
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